

Backed or Synthetic? Execution Mechanisms and Order-Flow Footprints in the Underlying Stock Market

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August 2026

Abstract

Claims on U.S. stocks trade continuously on crypto venues, whereas the underlying stocks trade only in scheduled sessions. On six such stocks over 2025–2026, creating a share-collateralized tracker token requires stock collateral but leaves the acquisition to the issuer’s discretion, whereas a cash-settled perpetual liquidation is forced when a margin threshold is crossed. Using event-level records of both, we find no detectable creation footprint, while forced liquidations coincide within minutes with elevated stock volume and directionally aligned order flow, and post-onset price movement stays below the move already in place. Voluntary pressure above the same dollar threshold leaves no detectable aligned footprint; a pre-onset match remains positive, with balanced observed states but imperfect range overlap. Because prices trigger liquidations, we report associations and cannot distinguish execution urgency from forced-fill visibility, remaining state selection, or common shocks.

Keywords: Tokenized equities; Perpetual futures; Order flow; Forced liquidation; Market microstructure

JEL classification: G13; G14; G23

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1. Introduction

Tokenized U.S. equities and synthetic perpetual futures on individual stocks trade continuously on crypto venues, while the underlying stocks trade during scheduled sessions. The underlying stocks trade during a regular session, thinner extended sessions, and a single small overnight venue, but not on weekends. This mismatch creates a directly testable market-structure question: when an obligation to trade arises in a market that never closes, is it accompanied by order flow in the underlying stock, and does the pattern vary with stock-market access and execution urgency?

The setting supplies two execution mechanisms that differ sharply in urgency while sharing the underlying names, sample period, and set of stock-side measures. The Backed (xStocks) and Ondo products are third-party-issued tracker certificates and debt instruments whose issuer obligations are collateralized one-for-one by shares held with custodians. Token holders own the certificates rather than the referenced shares or interests in a fund. We therefore use *share-collateralized* and *cash-settled perpetual* as operational labels, not legal classifications. Creating a tracker token requires corresponding stock collateral to be established, but it need not produce a contemporaneous purchase in a lit market; Backed's documentation permits in-kind delivery in which no purchase occurs. Cash-settled perpetual futures hold no shares, and when a levered position falls below its maintenance margin, the venue's engine closes it with market orders, immediately and without discretion. The tracker token creates a collateralization obligation and leaves the timing of any stock acquisition to the issuer. The perpetual creates no such obligation, but its engine forces trades that someone may need to offset at once. The comparison does not hold the issuer, venue, or event-generating process fixed, so execution

urgency remains a candidate explanation for the contrast rather than an isolated treatment.

We observe the mechanism-relevant events directly rather than inferring them from prices. The on-chain record contains every token creation and redemption with its transaction timestamp, and for the largest creations we trace the on-chain payment leg to narrow the possible settlement channels. The perpetual venue’s public archive contains every forced liquidation with price, size, direction, and account identifiers (roughly 669,000 fills, about 54,000 in our six names). We measure trading in the underlying stock at the minute level around these events, netting out each stock’s time-of-day pattern against matched control days and comparing liquidation windows with increasingly similar non-liquidation windows using only pre-onset information. When we detect no response, we report confidence bounds on its possible magnitude. The stock side is measured on a broad feed whose median share of the Nasdaq daily total is 79% on creation event days.

Large share-collateralized tracker-token creations, with a median value of \$2.4 million, are followed by no detectable change in the underlying stock’s volume, volatility, spreads, auction imbalance, or opening gap over the minute- and day-level horizons we examine. These horizons include the minutes around each creation’s actual timestamp and, for creations stamped overnight, trading on the only stock venue open at the time. We report confidence bounds rather than interpret this null result as evidence of no effect: point estimates for abnormal volume and signed order flow are 0.21 and -0.05 standard deviations, with one-sided 95% confidence bounds of 0.39 and 0.06, though smaller or temporally dispersed responses cannot be excluded. The settlement evidence identifies channels through which public-market trading could be attenuated. None of the 95 large creations covered by the payment-trace analysis shows a matching on-chain USDC payment, leaving in-kind delivery and off-chain cash settlement

observationally indistinguishable. The issuer's executing broker discloses that it routes order flow to wholesale market makers, and 42 to 57% of weekly volume in these names executes off-exchange.

Forced liquidations show a different pattern. When liquidations cluster on the perpetual venue during regular hours, volume in the underlying stock rises 1.68 standard deviations above its time-of-day norm over the eleven-minute window centered on the threshold bin ($t = 8.9$), and signed order flow tilts in the liquidation's direction by 1.09 standard deviations ($t = 4.1$). Both responses survive subtracting the elevated pre-event background. The pooled difference between the two event classes' volume responses is 1.47 standard deviations ($t = 6.9$). We also measure price movement after the cascade begins. We sequence the stock's ticks against the cascade's operational onset, defined as the first observed forced fill after at least thirty quiet minutes. The aligned price move is concentrated before the cascade begins—46 basis points before versus 10 after over the two minutes on each side ($t = 7.2$ and 1.4), and the one-sided bound on the post-onset move, 21 basis points, is under half the realized pre-onset move. We therefore detect an association in trading activity but no detectable incremental price impact beyond the move that triggered the event.

Because liquidations are triggered by price moves, the association could reflect a common shock rather than the event. We therefore add controls in progressively stronger steps. The raw footprint first survives subtraction of activity measured before the cascade's first observed fill. We then compare every burst with twenty non-liquidation windows of the same stock near the same time of day, first matching only the pre-onset stock return and then adding the pre-onset stock path, realized volatility, volume, absolute order imbalance, perpetual return and turnover, index-futures return, and funding. Under the stronger match and background adjustment, the

differences remain 1.10 standard deviations for volume ($t = 9.0$) and 0.92 for aligned flow ($t = 3.5$); excluding scheduled earnings dates and their preceding dates gives similar estimates. The trading schedule provides a separate diagnostic: after-hours, premarket, and overnight bursts coincide with elevated trading on the stock venue open at the time, while weekend bursts have no contemporaneous stock outcome to measure. Re-timing at the cascade onset shows that the stock's fill rate increases after the first observed forced fill, but its direction and most of its price movement are already present beforehand. Matching on observed market states and excluding scheduled announcements do not eliminate the footprint, but unobserved common shocks may remain. We therefore interpret the estimates as associations.

A second comparison holds the venue, instrument, dollar threshold, and merging rule fixed, while contrasting observed forced fills with voluntary net taker pressure. The size measures are related but not identical: the threshold applies to gross liquidation notional for forced bursts and absolute net taker flow for voluntary windows. Directional stock flow in voluntary windows is indistinguishable from zero, and the raw forced-minus-voluntary difference is 1.09 standard deviations ($t = 3.5$). We then pair each of the 39 forced bursts with a unique same-stock, same-direction voluntary window near the same time of day, using five state variables that end before the actual or pseudo-onset and excluding outcomes from the assignment. The matched aligned-flow difference is 0.77 standard deviations ($t = 2.7$), and the volume difference is 1.12 ($t = 4.5$). The largest paired standardized difference across the five matching variables is 0.10, but only 17 of 39 forced events lie within each of the five marginal candidate ranges. The contrast is therefore consistent with execution urgency as a candidate mechanism, but it does not distinguish urgency from forced-fill visibility, residual state selection, or common shocks.

The paper makes two contributions. First, we compare two execution mechanisms in a common event-time framework using the same set of underlying-market measures. The contrast is consistent with a role for execution urgency rather than collateral backing alone: forced liquidation coincides within minutes with directionally aligned trading, while a larger median collateralization obligation produces no detectable response within explicit confidence bounds. The latter result is also consistent with the settlement evidence. Second, we construct a same-threshold comparison of forced and voluntary perpetual pressure on the same venue and balance five observed pre-onset state and size dimensions in a one-to-one match. We also remove each liquidation event's pre-onset background, compare it with same-stock and same-clock low-liquidation windows matched only on pre-onset information, exclude scheduled-earnings dates, measure the resulting association in dollars, and re-time the sequence at the cascade onset. These steps reduce several observed-state explanations but do not isolate a causal effect or separate urgency from visibility. We cannot decompose how the tracker-token collateral was sourced across off-exchange purchases, internalization, pre-positioned inventory, and in-kind delivery, and our evidence comes from six names over a single year on one perpetual venue.

The paper builds on the market-microstructure literature on where and when trading occurs. Barclay and Hendershott (2003) document how thin off-hours trading is, which is why we measure the stock side session by session rather than in regular hours alone. The broader framing draws on venue competition (Chowdhry and Nanda, 1991) and constrained arbitrage (Gromb and Vayanos, 2002).

Studies of replicating products ask whether creation and redemption transmit non-fundamental demand to underlying assets (Ben-David et al., 2018; Da and Shive, 2018; Brown et al., 2021). The creation and redemption mechanism of a share-collateralized tracker

token is economically comparable to that of a physically backed vehicle, while the perpetual is a cash-settled derivative. This is a mechanism analogy, not a claim that the tracker tokens are fund interests. Unlike the index-level coexistence of exchange-traded funds and futures, both products reference the same single names and their primary events are recorded individually. The tracker-token channel leaves no footprint we can detect.

Bian et al. (2026) show that margin-driven selling moves prices within an equity market. Our liquidation results document a related pattern across the crypto–equity boundary, in trading activity rather than in additional price impact. The levered exchange-traded fund literature (Cheng and Madhavan, 2009; Shum et al., 2016; Ivanov and Lenkey, 2018) supplies the closest mechanical analog.

Recent studies examine price tracking and discovery in tokenized equities (Cong et al., 2025; Dombrowski and Arshadi, 2026; Mazur and Polyzos, 2026; Scharnowski, 2026). We instead use observed creations and forced liquidations to study whether execution obligations are accompanied by order flow in the underlying stock.

The remainder of the paper proceeds as follows. Section 2 describes the trading schedule, the two execution mechanisms, and the mark-price trigger for liquidation. Section 3 describes the data. Section 4 presents the event-level results for creations and liquidation bursts. Section 5 develops the pre-onset controls, sequencing evidence, and limits of interpretation. Section 6 concludes.

2. Market structure and trading mechanisms

2.1. Trading hours

The setting's defining feature is a mismatch in trading hours. Claims on U.S. stocks trade continuously on crypto venues, while the stocks themselves trade across several venues with scheduled closures. These schedules determine when a trade in one market can be met by a trade in the other.

The stock side runs as follows. The regular session is 9:30 to 16:00 Eastern time. Extended-hours trading continues until 20:00 and resumes at 04:00 the next morning. Between 20:00 and 04:00, the stock itself trades only on an overnight alternative trading system operated by Blue Ocean Technologies. The system runs Sunday through Thursday nights and serves mainly retail brokers routing overnight orders. Each session precedes the next trading day, so the venue does not open on the night before a market holiday. In our sample, the eight nights preceding a holiday contain no trades, compared with a full eight-hour session on ordinary nights. Its prints reach the consolidated tape the following morning rather than in real time, and its weekly volume in our names averages 0.28 to 0.78% of consolidated volume. The venue is small relative to the trading day but is the only stock venue available during its operating hours, which sharpens the session comparison in Section 5. Index futures trade nearly around the clock on weekdays but halt from Friday 17:00 to Sunday 18:00.

Stacking these schedules partitions the week. In the regular session, the stock, the tokens, and the perpetual trade simultaneously, and the stock price provides a common reference point. On weeknights, the stock trades only on the overnight system while the crypto products trade freely. From Friday 20:00 to Sunday 18:00, no stock venue and no index future operates, and the

crypto products are the only markets in which claims on these stocks trade. The tokens and the perpetual trade at all hours.

This partition serves two purposes in the empirical design. Weekend windows contain no stock venue and no index future, whereas weeknight windows can still contain trading on the overnight system, whose activity we observe and measure separately. For Sections 4 and 5, the schedule also varies the stock-market capacity available to a trader seeking to offset a position in the underlying. That capacity ranges from full regular-hours depth to one small venue at night and no stock venue on weekends, while the crypto-side mechanisms described next operate at all hours.

2.2. Share-collateralized tracker tokens: a discretionary collateralization obligation

The market label *tokenized equity* does not identify a single legal form.¹ The two lines in our sample are third-party-issued tracker certificates and debt instruments, not the referenced shares or interests in a fund. Backed Assets classifies each xStock as a bearer debt instrument and tracker certificate; Ondo’s approved prospectus likewise describes blockchain-based tracker certificates issued as debt instruments. Holders receive contractual economic exposure, while the corresponding shares collateralize the issuer’s obligation and are held one-for-one with custodians. Collateralization does not give the token holder direct shareholder title. We call these products *share-collateralized tracker tokens* to describe their economic backing, not to assign a regulatory category. Appendix B of the Supplementary Material records the product and

¹Tokenizing an existing security can involve native issuance, replacement of the legacy claim, or a custodian-issued claim on retained assets, and current token standards do not by themselves preserve shareholder rights when tokens move into smart contracts (Malinova and Park, 2026).

regulatory documents supporting this distinction.

Backed Assets issues the xStocks line, which trades on Solana-based decentralized exchanges and centralized crypto exchanges. Ondo Finance issues a second line of share-collateralized tracker tokens on the same underlying names. The two programs differ in how primary flow arrives. Backed issuance is intermediated: authorized participants submit creation orders that the issuer fills in batches, so large creations are infrequent and lumpy. Eligible customer purchases and sales in the Ondo program appear one-for-one as creations and redemptions, so its primary flow is small and nearly continuous. Section 4 measures footprints around the lumpy Backed creations, and the Supplementary Material reports the Ondo daily-flow regressions.

A creation requires corresponding share collateral to be established, but it does not necessarily trigger an immediate stock purchase or any purchase in a lit market. How much discretion the issuer holds is a property of the settlement mechanics, and those mechanics are what Section 4 tests. Backed's documentation describes two settlement paths. Under the market-flow path, the subscriber pays cash, the issuer's executing broker buys the shares, and the tokens are delivered once the shares settle; the issuer states that processing takes thirty seconds to two minutes. Under the in-kind path, the subscriber delivers the shares themselves and no purchase occurs.² Which path a given creation took is not disclosed, but every creation and redemption is recorded on a public blockchain with a transaction timestamp, and the payment leg, when it occurs on-chain, is visible in the same record. These timestamps and payment traces are the raw material for the event alignment and the payment-trace analysis in Section 4.

²That broker's own order-routing disclosures report that it routes customer order flow to wholesale market makers rather than to lit exchanges, and off-exchange executions account for 42 to 57% of weekly volume in the six stocks we study, so even market-flow purchases need not reach a lit exchange.

Creations and redemptions are processed around the clock on weekdays. The tokens themselves trade at all hours, weekends included.

2.3. Synthetic perpetual futures: a forced-execution mechanism

The synthetic products are perpetual futures on individual U.S. stocks, listed on a venue built on the Hyperliquid blockchain. A perpetual future is a cash-settled contract with no expiration date. The venue holds no shares, and neither do the traders; positions are collateralized with stablecoin margin and levered. Because the contract never settles into the underlying, its transaction price is formed in the venue’s order book and is economically linked to the stock through funding and arbitrage: at regular intervals, a premium to the reference price makes long positions pay short positions, and a discount reverses the payment. Ackerer et al. (2026) analyze how this funding rule anchors a perpetual to its reference asset. A stock-derived oracle is one input to the mark price that governs margining and liquidation; Section 2.4 describes this trigger.

The mechanism that matters for the underlying market is automatic liquidation. When adverse price moves push a levered account below its maintenance margin, the venue’s liquidation engine closes the position with market orders, immediately and without discretion. The position must be closed at the prevailing price and in the size that the account carries. This is the sharpest available contrast to the collateralization obligation of Section 2.2, which permits the issuer to choose when, where, and whether to trade in the lit market. Because the venue operates on a blockchain, every liquidation fill is archived publicly with its price, size, direction, and account identifier.³ Section 4 treats clusters of these fills as observable, precisely timed events of

³The archive spans roughly 669,000 fills across the venue’s listings, of which about 54,000 are in the six stocks we study.

forced order flow inside the crypto ecosystem and asks whether trading in the underlying stock accompanies them.

2.4. Margining and the liquidation trigger

The transaction price comes from the venue’s order book. For margining, the venue instead uses a mark price: the median of a stock-derived oracle, an oracle-plus-smoothed-basis component, and a component built from the best bid, best ask, and last trade. The mark price governs maintenance margin and triggers liquidation. Stock and perpetual price changes can therefore jointly precede a forced fill, making liquidation timing endogenous to the price process. This feature motivates the associational interpretation and pre-onset controls in Sections 4 and 5.

External stock inputs cover Sunday 20:00 to Friday 20:00 Eastern time, including the overnight system from 20:00 to 04:00. When those inputs are unavailable, the documented fallback begins from the last external price and adjusts using order-book impact prices. Weekend marks therefore retain a stale stock reference but receive no new contemporaneous stock input. We do not observe the oracle series itself and use the documentation only to characterize the liquidation trigger and the variation in stock-market access.

3. Data

3.1. Sample

The stock-outcome estimation sample covers the six U.S. stocks with the most active tokenized markets—Nvidia, Tesla, Circle, Coinbase, Alphabet, and Robinhood—from July 2025 through

June 2026. The product-launch census begins in June 2025 and contributes twelve core-name creations to the 112-event count in Table 1; those observations precede the stock minute panel and therefore do not enter any stock-footprint estimate. The tokenized equity market began trading at scale in mid-2025, so the estimation sample spans essentially the full period for which the stock outcomes can be constructed rather than a window selected from a longer history. Three panels start later than the stock data and therefore limit the specifications that use them: the perpetual contract was listed in October 2025; first records in the taker-signed all-fill panel used for the forced–voluntary comparison arrive between November 12 and December 15, 2025, depending on the stock; and market data for the overnight trading system is available only from August 24, 2025. For a stock whose first all-fill record arrives after the 10:00 ET analysis-band start, we discard that left-censored date and begin on the next equity session. Measures that require these panels begin at their stated coverage dates rather than July.

Among the nine single-name tokenized U.S. equities with continuous two-sided quotes on verifiable venues, these six carry 96% of token dollar volume and 87% of perpetual dollar volume, so they account for most verifiable trading in this market. We classify as verifiable the on-chain venues, where trades are signed transactions, and the major centralized exchanges that pass standard wash-trading screens. We exclude volume reported by smaller unregulated exchanges whose totals fail standard wash-trading screens (Cong et al., 2023). The Solana and Bybit records represent the same transferable Backed xStock claim, not separate issuer instruments; we aggregate their recorded dollar volume without pooling different issuers' claims.

3.2. Sources

The analysis draws on several groups of data; Appendices B and C of the Supplementary Material describe each source and the construction steps. For the stocks, we use tick-level trades with prevailing quotes, minute bars, and opening and closing auction imbalance feeds.⁴ The stock side is measured on XNAS.BASIC, which consolidates the Nasdaq-family exchanges and the two Nasdaq trade-reporting facilities. It is not the full securities information processor (SIP) tape.⁵

From XNAS.BASIC, we take minute bars for volume and variance measures, and trades with prevailing quotes for effective spreads, signed order flow, and the second-level sequencing in Section 5.3. Trades are classified relative to the contemporaneous best bid and offer (BBO) midpoint, with a within-trading-day tick-test fallback. Continuous one-minute feed-level consolidated quotes from the same source supply midpoints for the hours in which trades are sparse. The opening gap comes from the auction cross rather than a trade feed. Official opening and closing prices are taken from the statistics feed of each stock’s primary listing exchange: Nasdaq TotalView-ITCH for the five Nasdaq-listed names and NYSE Pillar for the NYSE-listed name. These records identify the completed auction cross and differ from the pre-cross reference and indicative clearing prices in the imbalance feed, which we use only to measure auction-imbalance pressure.

For the tokens, we observe Solana decentralized-exchange trades and Bybit five-minute

⁴The tick data additionally cover a three-minute window before each candidate control window of the matched design in Section 5.2, so that the final minute before a pseudo-onset is observed at the same resolution for events and controls.

⁵Its median share of the Nasdaq historical daily volume total ranges from 53% to 87% across the six names, and we do not observe executions on venues outside it. The share is stable across event and non-event days: within stock, coverage on liquidation-event days differs from the stock’s median by -0.4 percentage points ($t = -0.6$), so shifts in venue routing do not account for the abnormal-volume estimates, and the small deficit on event days works against finding elevated volume.

bars. The perpetual data come from one cash-settled contract venue, whose public archive provides five-minute and one-second bars, the funding-rate history, liquidation fills, and per-fill records with the aggressing side flagged, from which we construct taker-signed net flow for the forced-versus-voluntary comparison. The perpetual results therefore describe that contract and venue, not a consolidated perpetual market. For the overnight stock venue, we use minute bars, quotes, and trade prints. The on-chain record provides the full census of token creations and redemptions across the four blockchains on which the issuers operate, including transaction timestamps and counterparty wallets. For the largest creations, we also trace the on-chain payment leg to narrow the possible settlement channels. Supplementary creation tests additionally use five-minute index-futures, bitcoin, and ether returns, the VIX, an earnings calendar built from time-stamped regulatory filings, and ex-dividend dates.

Our design differs from return-based studies of the same market. Creations and redemptions identify primary issuance events; liquidation fills identify forced executions; and aggressor flags identify voluntary taker pressure. These measures record when an execution obligation or observed pressure arises and in what size, rather than only its price consequences. Because both the token registries and the perpetual venue operate on public blockchains, these records are event-level rather than sampled reports, which limits reporting-selection concerns.

3.3. Construction and measurement filters

Return windows are built with previous-tick boundary prices: the price assigned to a window boundary is the last transaction at or before that instant, never an interpolation, so that no boundary price is imputed. Daily variables are assembled into a stock-day panel of roughly 1,500

observations.⁶

Quote data require one consequential filter. Outside regular hours, some quotes are placeholder postings with spreads of tens of percent, and midpoints computed from them can sit far from any transaction price. We therefore discard any quote whose relative spread exceeds 5% before computing midpoints. These rules are stricter in our data than the spread-scaled cleaning rule of Barndorff-Nielsen et al. (2009) and the neighborhood-dispersion filter of Brownlees and Gallo (2006); measures computed under their original rules lead to the same conclusions.

Table 1 reports the event samples.

[Insert Table 1 approximately here]

4. Event-level footprints in the underlying market

The two obligations differ sharply in execution urgency. A share-collateralized tracker-token creation requires corresponding stock collateral but leaves the timing, venue, and even the necessity of a purchase to the issuer's discretion. A forced liquidation is executed by an engine at the prevailing price, in whatever size the position carries, when a margin threshold is crossed. This section measures the stock-market activity around each event using a common event-time framework and the same outcome measures.

Throughout, we measure trading activity: abnormal volume and the direction of order flow in the underlying stock. Section 4.2 shows that the liquidation-side association does not extend to detectable price impact beyond the move that preceded the event.

⁶The panel carries realized variance, bipower variation, and the implied jump share computed from minute returns following Barndorff-Nielsen and Shephard (2004, 2006), alongside effective spreads, signed order flow from quote-based trade signing, auction imbalance, and opening gaps.

Let $y_{i,e,\tau}$ denote the raw minute outcome for stock i at event time τ of event e , measured in log dollar volume or in signed order flow aligned with the event direction. We standardize each minute against the same stock at the same minute of the day on control days, and average the standardized values over the event window,

$$x_{i,e,\tau} = \frac{y_{i,e,\tau} - \mu_{i,m(e,\tau)}}{\sigma_{i,m(e,\tau)}}, \quad X_e = \frac{1}{11} \sum_{\tau=-5}^5 x_{i,e,\tau}, \quad (1)$$

where μ and σ are the mean and standard deviation of y at minute-of-day $m(e, \tau)$ over up to twenty control days within fifteen sessions of the event, excluding days that carry an event of the same type. The window statistic X_e is therefore expressed in control-day standard deviations, and a purchase concentrated in any single minute enters it at one eleventh of its size. Event-level comparisons take the form

$$X_e = a + b D_e + c' W_e + u_e, \quad (2)$$

in which D_e marks the event class being contrasted, such as a liquidation burst against a creation in Section 4.2 or a forced against a voluntary window in Section 4.3, and W_e collects pre-window state variables where stated. We estimate b by ordinary least squares with observations weighted equally across events, and we cluster standard errors on the calendar date of the event, with degrees of freedom set to the number of date clusters less one. Dates rather than stocks are the clustering unit because events in different names arrive together when the crypto market moves, and the date count exceeds the number of names by an order of magnitude.

4.1. Share-collateralized tracker-token creations

The prediction for a share-collateralized creation is straightforward: a large creation requires corresponding stock collateral, so if that collateral is obtained through a time-compressed public-market purchase, the purchase should be visible in trading around the event. The exchange-traded fund evidence closest to this mechanism, which links creation and redemption to demand for underlying stocks (Ben-David et al., 2018), would lead one to expect a footprint under that settlement path. The legal forms are different: the tokens in our sample are tracker certificates rather than fund interests. We detect no footprint in the observed public-market outcomes, and the rest of this subsection quantifies the bound and examines settlement channels consistent with it.

Timing matters because creations respond to demand. A creation recorded during a busy trading day would correlate with volume even if market activity caused the issuance rather than the reverse. We therefore begin with the localized design in Table 2, Panel A. Each event minute is compared with the same stock and minute of the day on nearby non-event days, and the second estimate also subtracts the event's own pre-event background. Creations recorded during regular hours (10:00 to 15:30 ET; the half hours adjacent to the opening and closing auctions are excluded throughout) show no abnormal signed flow in the event window of Eq. (1), at -0.05 standard deviations ($t = -0.8$), and no increment over the pre-event background (-0.06 , $t = -0.9$). Window dollar volume is mildly elevated (0.21 , $t = 1.99$, $p = 0.06$), but the same elevation is present through the half hour before the mint and the increment over that background is zero (-0.00 , $t = -0.03$), a pattern consistent with creations being timed to already-active periods rather than generating trading.

Panel B extends the same event-time comparison to the stock venue open at an off-hours timestamp. Off-hours creations are roughly two-fifths of the census. The eleven stamped overnight leave flat volume on the overnight system (-0.32 standard deviations, $t = -0.8$). Premarket creations occur during mildly elevated volume ($+0.33$, $t = 4.5$), but that elevation is present from thirty minutes before the event through thirty minutes after it. The increment at the event is essentially zero, a pattern more consistent with busy mornings generating token demand and creations than with creations generating stock trading.

A second concern is the ordering: an issuer that holds the shares before it mints would trade ahead of the timestamp, where a window centered on the mint would not look. We therefore apply the same construction to successive thirty-minute intervals extending backward—from thirty minutes before the mint through the same day's open and then through the previous trading day, so that a Monday mint is compared with the preceding Friday—in thirty-minute intervals, computing the abnormal z of each against the same stock's same clock interval on control days (Supplementary Material IA.8). Buy-aligned flow shows no consistent direction across the fifty-two intervals: the mean abnormal flow is -0.01 , and the few intervals that reach a t -statistic of two are as often negative as positive. Trading is somewhat elevated over the same span, at a mean abnormal volume of 0.23 , but it carries no direction. Concentrated purchasing in any observable interval of the preceding trading day would appear here; purchases made overnight, when the feed does not observe, or spread evenly across the day would not, and we cannot exclude them.

Outcomes measured over the opening hour, reported in Supplementary Material IA.7, likewise show no response on any of the eight outcomes; the one nominally significant coefficient, the jump share ($t = 2.14$), carries a max- t adjusted permutation p of 0.23 across the

eight. We do not read these opening-hour results as evidence either way.⁷ The tests that carry power here are the ones measured over minutes.

Sample counts differ across designs for identifiable reasons.⁸ Economically meaningful redemptions are effectively absent. Excluding negligibly small operational transfers, total burns amount to 0.0001 to 0.0011% of gross issuance in the non-crypto names, so the event analysis is confined to creations. Across the sample the 112 creations sum to \$0.65 billion, or 0.005% of the six stocks' regular-hours dollar volume over the same period, so the collateral channel is small in aggregate whatever form its execution takes.

A null of this kind raises the question of statistical power, which we answer by converting event size into the units in which the estimates are reported. The median large creation is roughly 3% to 4% of observed five-minute stock volume at the same time of day.⁹ Against a one-to-one purchase of the creation's own size, the volume test therefore has little power. It is not so limited against the alternative that a creation sets off the wider trading that accompanies a liquidation burst, which reaches 1.68 standard deviations in the same units from events carrying a fifth the notional; the pooled difference between the two event types, reported in Section 4.2, is 1.47 standard deviations with a confidence interval of 1.05 to 1.89. Signed order flow speaks to the narrower question of whether the collateral itself was bought in the open market, and there the test operates close to its detection threshold: the flow dispersion has a median of \$3.0 million per

⁷A creation of median size is a fifth of a percent of the opening hour's dollar volume in the same stock, against a standard error on signed order flow of 1.31%, so a purchase of that size would move the estimate by a small fraction of one standard error whatever form it took.

⁸The on-chain census contains 112 large creations in the six names (Table 1); the USDC payment trace, which follows USDC alone, covers 95 large mints in the core symbols; the minute design covers the 61 creations stamped between 10:00 and 15:30 ET, which collapse to 59 distinct stock-minutes and then to 47 windows with complete outcome and control coverage; five creations stamped in the half hours adjacent to the auctions enter neither the minute design nor the off-hours design; and the power calculation uses the 47 complete windows.

⁹Because volume enters in logs scaled by a control dispersion of 0.53, a purchase of that size adds only about 0.06 standard deviations to the window mean, far below the volume bound of 0.39.

minute, so a median creation of \$2.1 million in the opening-hour sample bought entirely inside the window would raise the eleven-minute mean by about 0.06 standard deviations, approximately the one-sided 95% bound of 0.06 (estimate -0.05 , $t = -0.8$). Liquidation-scale responses are excluded and substantially larger creations would register, but a one-to-one public-market purchase of median size lies at the edge of the confidence band rather than inside the rejection region, and purchases dispersed over hours or executed off the observed venues would not register at all. The feed covers 79% of the daily Nasdaq total on the 47 creation windows in the estimation sample.

A related objection is that one-minute aggregation could miss a purchase placed and filled within seconds. We therefore repeat the second-level design of Section 5.3 at the creation timestamp for the 45 event windows with complete tick data. Over the two minutes after creation, the buy-aligned flow tilt is -0.03 ($t = -1.1$), with a one-sided 95% bound of 0.018. The change in tilt across the timestamp is -0.04 ($t = -1.3$), and fills arrive at the same rate before and after (ratio 0.99, bound 1.08; Supplementary Material IA.5). The corresponding liquidation values, a tilt of 0.10 and a rate ratio of 1.20, lie outside these bounds. A second-level creation response as large as the liquidation response would therefore have been detected. Smaller responses remain admissible. One of the eight second-level statistics is positive: the purchase-aligned mid return over the two minutes after the timestamp is 5.1 basis points ($t = 2.45$). We do not read a single estimate among eight as a footprint, since neither directional flow nor fill arrival changes across the same timestamp. The 39 liquidation bursts also produce minute-level volume and aligned-flow responses of 1.68 ($t = 8.9$) and 1.09 standard deviations ($t = 4.1$). Acquisitions executed outside the venues represented in XNAS.BASIC would not register, so the confidence bounds apply to localized public-market activity rather than every

possible acquisition channel.

The settlement records narrow, but do not identify, the possible acquisition channels. For the 95 large creations covered by the payment-trace analysis, we test whether a matching on-chain USDC payment into the issuer’s operating wallets precedes the event. None is found, and the median on-chain inflow near a creation is \$170, compared with a median creation of \$2.4 million. The absence of an on-chain payment leg leaves in-kind delivery and off-chain cash settlement observationally indistinguishable. Issuer documentation permits both, including in-kind delivery in which no purchase occurs. The issuer’s executing broker discloses that it routes order flow to wholesalers, and 42 to 57% of weekly volume in these names executes off-exchange. A four-hour window designed to capture orders executed gradually also shows no accumulation of buying (Supplementary Material IA.2).

Abnormal trading volume over the one to four trading weeks after a creation is also indistinguishable from zero, providing no evidence of delayed accumulation (Supplementary Material IA.4). Our design cannot evaluate collateral acquisitions placed days before a creation. Because issuance follows demand, pre-creation trading cannot be separated from the demand that prompted the issuance. We therefore confine the claim to the observable event window, which covers the issuer’s disclosed processing time of thirty seconds to two minutes for the market-flow path. In our sample, share-collateralized creation obligations leave no detectable localized footprint in the public-market outcomes we observe around those events. The data do not identify how collateral was sourced across off-exchange purchases, internalization, pre-positioned inventory, and in-kind delivery.

[Insert Table 2 approximately here]

4.2. Synthetic-side liquidation bursts

The synthetic channel's events are forced liquidations. When clustered liquidations push levered positions into the engine, the resulting fills are market orders within the crypto market; the contract itself requires no trade in the stock. We ask whether stock-market trading nevertheless accompanies these fills through the hedging and arbitrage activity that connects the two markets. Bian et al. (2026) document, within an equity market, that margin investors liquidate more heavily as their accounts approach a margin call. Our setting asks whether a related pattern appears across the crypto–equity boundary, where intermediaries rather than direct sales of the underlying connect the markets.

An event is a five-minute window in which liquidation notional on the perpetual venue reaches at least \$250,000 during the same regular-hours span, merging qualifying bins within thirty minutes. This rule yields 39 events across all six names, with a median liquidation notional of \$430,000. The median burst is about a fifth the notional of the median creation, although only the creation carries a collateral obligation. Outcomes are measured as in the creation study, relative to the same minutes for the same stock on control days, after excluding other event days. Table 3 and Fig. 1 place the liquidation and creation estimates side by side. Pooling the two event samples in a regression of abnormal volume on a liquidation indicator yields a difference of 1.47 standard deviations, with a date-clustered t -statistic of 6.9 and a 95% confidence interval of 1.05 to 1.89, and a corresponding aligned-flow difference of 1.14 ($t = 4.1$); Table 3 reports both. Both event classes are measured over the same eleven-minute window and the same abnormal- z construction, so the two responses are measured identically; the contrast is between two measured responses and is not an estimate of what liquidation does to the stock.

The five-minute bin is a measurement interval, not an estimate of the mechanism's structural delay. Holding the 39 events fixed, the abnormal-volume estimate ranges from 1.42 to 1.82 standard deviations across two-, ten-, and fifteen-minute outcome windows (minimum $t = 8.06$); aligned flow ranges from 0.64 to 1.54 (minimum $t = 3.61$). Shifting the start of each five-minute bin by one to four minutes produces similarly narrow ranges: 1.65–1.70 for volume and 0.94–1.19 for aligned flow. The fixed \$250,000 cutoff is also an operational rule. Lowering it to \$100,000 retains all six names and yields 94 events with the same qualitative pattern, whereas cutoffs of \$500,000 and \$1 million leave only 14 and 4 events. Supplementary Material IA.1 reports the full audit.

Over the eleven-minute outcome window centered on a burst's threshold bin, stock volume rises 1.68 standard deviations above its time-of-day norm ($t = 8.9$), and signed order flow in the same feed tilts in the direction of the liquidation, selling after long liquidations and buying after short liquidations, by 1.09 standard deviations ($t = 4.1$). The volume response survives subtracting the pre-event background, measured over the 25 minutes ending five minutes before each cascade's onset so that no forced fill has yet occurred (increment 1.06, $t = 8.9$), as does signed flow (0.93, $t = 3.7$). Date-level sign-flip p -values for the raw and background-adjusted aligned-flow estimates are 0.0001 in 9,999 draws, and leaving out any one stock leaves t -statistics of at least 3.1 and 3.0, and leaving out any one event leaves at least 3.8 and 3.4. We treat this as a minutes-wide association rather than a causal discontinuity; Section 5 compares it with increasingly similar non-liquidation windows using only pre-onset information.

The principal identification concern is that liquidations are endogenous to the price process itself. Common information and order flow can move both the stock and the perpetual; the stock-derived oracle then enters, alongside smoothed-basis and order-book or trade

components, the mark price that governs maintenance margin and liquidation. A naive reading would therefore attribute the surrounding turbulence to the liquidation rather than to whatever moved prices and pushed the mark through the margin threshold. Liquidations against the direction of the prevailing move are rare in the data (2 of 94 candidate events). The design therefore leans on comparisons rather than a causal reading of timing: background subtraction removes turbulence already in progress, date-level sign-flip tests avoid treating multiple same-day events as independent evidence, and the matched windows of Section 5.2 ask whether comparable pre-onset market states without forced liquidations have the same stock footprint.

The interpretation is limited to trading activity rather than additional price impact. We therefore sequence stock ticks around the cascade’s operational onset—the first observed liquidation fill after at least thirty minutes without another fill—rather than around the later threshold-crossing bin. Complete two-sided windows cover 35 events on 32 dates. Over the two minutes before onset, the liquidation-aligned stock return is 46.3 basis points ($t = 7.2$), compared with 9.5 basis points after onset ($t = 1.4$; one-sided 95% bounds of 21 basis points at two minutes and 41 basis points at five, against realized pre-onset moves of 46 and 76). Stock fills arrive 1.20 times as fast after onset ($t = 2.7$), but their directional tilt is already 0.10 before onset and 0.09 after it; the change is -0.01 ($t = -0.5$). Thus, the price move is concentrated before the first observed forced fill, whereas trading becomes faster without a detectable change in direction afterward. This sequence does not identify which traders submit the aligned orders and provides no evidence that liquidations add price impact beyond the preceding move.

A common-shock alternative remains: the information that pushes the mark through a maintenance-margin threshold could also affect stock-market trading. Section 5 therefore adds controls in a fixed sequence rather than treating event timing as quasi-exogenous. The raw

footprint is first adjusted for the event's own pre-onset background. Each burst is then compared with twenty clean windows in the same stock and within two hours of the same time of day. A minimal comparison matches the thirty-minute stock return ending before onset; a stronger comparison additionally matches pre-onset stock returns, realized volatility, volume, absolute order imbalance, the prior five-day stock return, the stock's tick-level return over the final minute before onset, perpetual return and turnover, index-futures return, and funding. Every covariate ends before the actual onset, and each control receives the treated event's onset-to-threshold lag. All 39 bursts are matched, with a largest absolute paired standardized difference of 0.29.

Under the stronger match and background adjustment, the volume difference is 1.10 standard deviations ($t = 9.0$) and the aligned-flow difference is 0.92 ($t = 3.5$). Excluding eight events on a scheduled earnings date or the preceding date gives 1.13 ($t = 7.7$) and 0.96 ($t = 2.9$). The corresponding mean dollar differences are \$171.1 million in trading volume ($t = 3.4$) and \$31.5 million in liquidation-aligned net flow ($t = 2.4$), equal in aggregate to 89.2 and 16.4% of normal same-clock volume. These amounts describe the economic magnitude of the conditional association, not dollars caused by a dollar of liquidation. The design accounts for scheduled announcements and observed pre-onset market conditions, but not unscheduled news or other common shocks.

The scope of the result is bounded by two further checks. Three of the six stocks have crypto-linked business models, so a common crypto shock could move both the liquidations and the stock. Restricting the sample to the three names without such a linkage leaves estimates of 1.65 for volume ($t = 6.4$) and 1.25 for aligned flow ($t = 4.9$). A relative-volume event rule cannot support inference across all six stocks: when stock depth is measured over 10:00 to 15:30, a ten-percent cutoff leaves six CRCL events and a twenty-five-percent cutoff leaves one event. We

report these counts only to show the available sample, not as robustness estimates (Supplementary Material IA.1).

[Insert Table 3 approximately here]

[Insert Figure 1 approximately here]

4.3. What the direction alignment does and does not identify

Direction alignment identifies the sign of the flow but not the traders behind it. A market maker that absorbs liquidated positions and rebalances a stock hedge could generate the pattern, consistent with the option-hedging evidence of Ni et al. (2005); an arbitrageur trading a gap between the perpetual and the oracle could generate it as well. Inventory costs may strengthen the incentive to rebalance quickly when only the overnight stock venue is open: Eaton, Shkilko, and Werner (2025, working paper) find that average realized spreads in their frequently traded nocturnal sample turn negative after sixty seconds, but they do not observe provider-specific holding periods. These are candidate channels, not identified traders. Our claim remains associational: forced deleveraging on the synthetic side coincides within minutes with directionally aligned order flow in the underlying stock. Stock-specific news arriving in the same minute as the burst remains an unresolved alternative.

The following diagnostics narrow the set of candidate channels without selecting one. First, the scale of the association is far from the notional that triggers it: the median burst carries \$430,000 of forced fills, while the matched excess is \$171.1 million of trading volume and \$31.5 million of aligned net flow per event. These are conditional associations rather than multipliers, but their size implies that a one-for-one hedge of the liquidated notional cannot account for the

flow we measure, although it does not exclude such hedging as a component. Second, the alignment is two-sided (Table 4, Panel A). Stock flow tilts toward selling after long liquidations (0.79 standard deviations, $t = 3.4$; 27 events) and toward buying after short liquidations (1.75, $t = 2.8$; 12 events on 10 days). Explanations that operate only in declines, such as margin-account selling in the stock itself, are inconsistent with the buy-side response, though the short-liquidation sample is small. Third, the footprint is not explained by multi-day momentum states: regressing per-event increments on the prior five-day return gives slopes indistinguishable from zero (-0.070 per 1,000 basis points for volume, $t = -0.8$; $+0.227$ for aligned flow, $t = +0.8$; 39 events on 34 dates), and Section 5 shows that adding that return and the final pre-onset minute to the match leaves the estimates unchanged. Fourth, simultaneous multi-name bursts are uncommon: only four of thirty-four event days contain bursts in more than one name. This does not rule out heterogeneous responses to common news.

A sharper diagnostic compares forced fills with voluntary taker-pressure windows on the same venue that clear the same dollar threshold. The venue's fill records identify the aggressing side of every trade, so voluntary pressure requires no inference. First records in the taker-flow panel arrive between November 12 and December 15, 2025, depending on the stock. If a stock's first record arrives after the 10:00 ET analysis-band start, we remove that left-censored date and start on the next equity session; all samples end on June 30, 2026. We first retain qualifying five-minute bins inside each actual U.S. equity session, including holiday closures and early closes; require absolute taker-signed net flow of at least \$250,000; merge qualifying bins within thirty minutes; and then exclude windows with at least \$10,000 of liquidations within thirty minutes and stock-days containing a liquidation burst. We align the stock-flow outcome with the sign of the perpetual pressure. This sequence defines 876 windows, of which 470 on 137 dates

carry complete eleven-minute outcomes and 25-minute backgrounds; the remaining definitions fail stock-side outcome or control-day completeness. The threshold is common, but the two size measures are not the same economic object: among complete windows the median voluntary observation carries \$360,000 of absolute net taker flow and \$560,000 of gross taker turnover, while the median forced burst carries \$430,000 of gross liquidation notional.

[Insert Table 4 approximately here]

Table 4 reports the raw and matched comparisons. Voluntary windows show mildly elevated stock volume (0.53 standard deviations, $t = 8.9$) but no detectable pressure-aligned stock flow (-0.002 , $t = -0.01$). The raw forced-minus-voluntary differences are 1.15 standard deviations of volume ($t = 5.9$) and 1.09 of aligned flow ($t = 3.5$). We then form a one-to-one comparison that uses no outcome information. Each forced burst is paired with a unique voluntary window in the same stock, with the same pressure direction, and within two hours of the same clock time. Five matching dimensions are measured before the observed forced onset or the control's pseudo-onset: the aligned thirty-minute stock return, realized volatility, log stock dollar volume, absolute stock imbalance, and log trigger size. Forced-event states end before the observed cascade onset. For each possible control, the treated event's nonnegative onset lead is applied as a pseudo-onset, so its state also ends before the qualifying voluntary bin.

The assignment requires the matched-control mean of every dimension to lie within 0.10 reference standard deviations of the forced mean and then minimizes equal-weight standardized distance, using each control at most once. All 39 forced bursts are retained and matched to 39 unique voluntary windows. The largest ordinary paired standardized difference is 0.097. Inference uses one forced-minus-control difference per pair and two-way clustering by the forced

and control dates, which number 34 and 31. The matched differences are 1.12 standard deviations of volume ($t = 4.5$) and 0.77 of aligned flow ($t = 2.7$). Overlap is nevertheless incomplete: only 17 of the 39 forced states lie within each of the five marginal candidate ranges. The positive conditional contrast is consistent with execution urgency, but it does not isolate compulsion or distinguish urgency from forced-fill visibility, remaining state selection, or common shocks. Supplementary Material IA.9 reports the construction, balance, support, and background-adjusted sensitivity.

These patterns connect to prior evidence on trading that exceeds its trigger. Barbon et al. (2019) document that fire-sale information leaking through brokers draws third-party trading alongside the liquidating fund, although in their setting the additional trading raises price impact, which we do not detect here. Brunnermeier and Pedersen (2009) model funding spirals in which a shock forces many constrained traders to adjust simultaneously; if that mechanism operates here, the volume observed at a stress moment would reflect the capital of all constrained participants rather than the initiating position. Bian et al. (2026) provide account-level evidence of the same logic in stock margin accounts, where accounts near their limits sell together. Each mechanism generates responses untethered to the trigger's size, and each is consistent with parts of our pattern; our data do not separate them from one another, nor from a common price acceleration that both trips the engine and draws stock-market trading.

This association has a counterpart in the literature on levered exchange-traded funds. Their issuers rebalance mechanically toward a constant exposure near each day's close, buying after gains and selling after losses in the direction of the underlying's move (Cheng and Madhavan, 2009). Whether that rebalancing measurably reaches the underlying remains unsettled. The demand raises late-day volatility only on the most active days in Shum et al.

(2016) and is economically small on average once fund creations and redemptions absorb much of it in Ivanov and Lenkey (2018). Bessembinder (2026, working paper) is adjacent to this question: the paper studies investor returns for single-stock levered funds, decomposing return shortfalls into daily-rebalancing and friction components, rather than measuring their underlying-market footprint.

A forced liquidation differs from scheduled rebalancing in two observable respects. It occurs at an endogenous margin threshold rather than at a scheduled close, and we observe individual forced fills rather than infer demand from return decompositions. Unlike a levered fund rebalance, it is not mechanically paired with an issuer creation or redemption process. Other traders can nevertheless take the opposite side, and our data do not reveal whether subsequent stock trades are hedges, arbitrage, or responses to a common shock. We therefore treat the estimates as complementary evidence on mechanical trading, not as proof that an offsetting channel is absent.

5. A sequence of controls for the liquidation footprint

The event-time association in Section 4.2 is not a causal estimate. Liquidations occur after adverse price movements, so common information may generate both the forced fills and the surrounding stock-market activity. Table 3 therefore adds controls in a predetermined sequence. Panel A describes the footprint. Panel B removes activity already present before the first observed forced fill. Panel C compares each event with increasingly similar non-liquidation windows using only information available before that fill. Panels D and E report economic magnitude and event-time sequencing. Throughout, the quantity of interest is the difference in

stock trading around a liquidation burst after conditioning on progressively richer observed information. None of these steps randomizes the burst.

5.1. Description and within-event adjustment

The unadjusted threshold-window outcomes are 1.68 standard deviations of abnormal volume ($t = 8.9$) and 1.09 of liquidation-aligned flow ($t = 4.1$). These estimates document the raw pattern but do little to separate it from an already turbulent market. The first adjustment subtracts the mean outcome over the twenty-five minutes ending five minutes before the cascade's operational onset, defined as the first observed liquidation fill after at least thirty minutes without another fill. The outcome window remains centered on the threshold bin, where the event rule fixes the required liquidation volume; only the comparison period moves to a point before any observed fill in the cascade. The adjusted estimates are 1.06 ($t = 8.9$) for volume and 0.93 ($t = 3.7$) for aligned flow. This is a within-event comparison, not a counterfactual: a common shock may already be developing during the background period or intensify afterward.

The first liquidation and the threshold used to define an event do not always coincide. In 28 of 39 events, smaller forced fills begin before the five-minute bin first exceeds \$250,000. A background window placed immediately before that bin can therefore subtract the early part of the cascade. We retain that threshold-based background as a conservative diagnostic in the Supplementary Material, but use the pre-onset background in the main table because the latter ends before the first observed forced fill.

5.2. Matched non-liquidation windows

The next two rows add external comparison windows. Candidate controls come from the same stock, lie within two hours of the event's time of day, occur on non-burst and non-degraded days, and contain less than \$10,000 of liquidations in the surrounding thirty minutes. We assign each candidate the treated event's onset-to-threshold lag. Matching variables are measured up to the resulting pseudo-onset, while outcomes are measured around the corresponding pseudo-threshold. This construction aligns the outcome timing without allowing any matching variable to use information observed after liquidation begins.

The minimal design matches only the sign and nearest magnitude of the stock's thirty-minute return ending before onset. It asks whether the footprint survives the most direct observable confounder: a similar stock-price path before forced selling starts. Combining this match with the event's own pre-onset background leaves differences of 1.08 standard deviations for volume ($t = 8.7$) and 1.05 for aligned flow ($t = 4.0$).

The stronger design widens the match from the return path alone to the pre-onset market state. Eleven variables enter: the stock's five-minute and thirty-minute returns, thirty-minute realized volatility, dollar volume, absolute order imbalance, and cumulative return over the five trading days ending at the prior close; the stock's tick-level return over the sixty seconds ending one second before onset; the perpetual's thirty-minute return and turnover; the equity-index futures return; and funding.¹⁰ Each variable is standardized by its dispersion across the candidate pool, and each event receives the twenty candidates closest in root-mean-square distance across

¹⁰The five-day return is included because bursts sit inside multi-day directional runs. The median long-liquidation event follows a five-day decline of 321 basis points, so controls drawn from calm weeks would otherwise differ in trend state.

the variables it has observed.¹¹

All 39 events obtain twenty controls, and across the eleven pre-onset variables the largest absolute paired standardized difference is 0.29. The minute bars that supply the other covariates cannot see inside the minute containing the onset, which would leave roughly the final ninety seconds before the first forced fill unmatched; the tick-level variable reduces that gap to approximately one second for most windows, and the median absolute event-to-control difference on it is 7 basis points.¹² The five-day return is the one variable whose support is imperfect.¹³ Restricting to the 27 events whose controls are also close on the five-day return gives 1.16 for volume and 0.96 for aligned flow, both at least as large as the full-sample estimates. The matched level differences are 1.60 ($t = 8.7$) for volume and 0.97 ($t = 3.6$) for aligned flow. After also subtracting each window's pre-onset background, the main estimates are 1.10 ($t = 9.0$) and 0.92 ($t = 3.5$). Date-level sign-flip p -values based on 9,999 draws are at most 0.0002. The estimates are close to the minimal design rather than emerging only after the richer match.

Scheduled earnings are the most readily observed stock-specific news events. Eight bursts occur on an earnings date or the preceding date. Excluding them leaves 31 events on 28 dates and estimates of 1.13 for volume ($t = 7.7$) and 0.96 for aligned flow ($t = 2.9$). This screen cannot remove unscheduled corporate news, macroeconomic releases with heterogeneous stock exposure, or latent order-flow shocks. The matched estimates therefore show that controlling for the measured pre-onset return path and market state does not eliminate the footprint. They do not show what the footprint would have been had the liquidation engine not acted.

¹¹Directional order flow enters the match in absolute value rather than in liquidation-aligned form, so the eventual liquidation direction cannot leak into control selection.

¹²The last recorded quote precedes the cutoff by a median of 0.13 seconds and by about one second at the ninetieth percentile, with rare gaps of up to half a minute when trading pauses.

¹³A few events sit in stock-specific runs with no comparable week elsewhere in the sample, so their controls match the remaining state variables but not the trend.

5.3. Economic scale, sequencing, and the claim boundary

Applying the same strong match and background adjustment to dollar outcomes yields mean excess trading of \$171.1 million per event ($t = 3.4$) and liquidation-aligned abnormal net flow of \$31.5 million ($t = 2.4$). Summed across events, these amounts equal 89.2 and 16.4% of normal same-stock, same-clock dollar volume. The estimates remain positive when any one stock is omitted, although the least favorable omission lowers the aligned-flow t -statistic to 2.0. These amounts show that the conditional association is economically large, but they are not multipliers on liquidation notional: common news, hedging, arbitrage, and other trading can all contribute to the measured dollars.

The onset sequence further limits the mechanism claim. In the 35 events with complete two-sided tick windows, the liquidation-aligned stock return is 46.3 basis points before the first observed forced fill ($t = 7.2$) and 9.5 basis points afterward ($t = 1.4$), with a one-sided 95% bound of 21 basis points. Widening each side to five minutes preserves the ordering at larger magnitudes, 76.5 basis points before and 21.7 after ($t = 6.4$ and 1.9), so the post-onset movement grows with the window but stays a minority of the move already in place. The stock's fill rate rises by 20% after onset ($t = 2.7$), while its directional tilt changes by -0.01 ($t = -0.5$). Thus elevated, directionally aligned trading accompanies liquidation bursts after rich observable adjustment, and trading becomes faster when the observed cascade begins; the direction and most of the price movement are already present beforehand. The evidence is consistent with execution urgency and cross-market hedging as candidate mechanisms, but it does not establish that liquidations cause the stock flow or amplify the stock price move.

Supplementary Material IA.1, IA.3, and IA.6 report diagnostics for the window length,

bin origin, event cutoff, stock omissions, covariate balance, and trading session. Bursts in after-hours, premarket, and overnight sessions are accompanied by elevated trading on the stock venue open at the time, while weekend events have no contemporaneous stock outcome to measure. These checks show that the pattern appears across observed sessions and venues. They do not provide exogenous treatment variation and therefore remain outside the main sequence of controls.

6. Conclusion

Two execution mechanisms created claims on the same six stocks over the same year: one establishes a share-collateralization obligation while leaving any stock acquisition to the issuer's discretion, and the other forces immediate liquidation at the prevailing price. Forced liquidations coincide within minutes with elevated volume and directionally aligned order flow in whichever stock session is open, while share-collateralized creations leave no footprint our data can detect, with settlement records consistent with collateral sourced off-exchange, from prior inventory, or in kind. Voluntary pressure above the same dollar threshold leaves no detectable aligned footprint either. A one-to-one pre-onset match balances five observed state and size dimensions and retains a positive forced-minus-voluntary difference, but candidate-range overlap remains imperfect. The contrast is consistent with execution urgency; it does not identify urgency separately from forced-fill visibility, residual state selection, or common shocks.

Five limits apply. Liquidations are triggered endogenously by prices, so we report associations rather than causal effects. We do not observe the traders behind the aligned stock flow. We cannot decompose how tracker-token collateral was sourced. Unscheduled news inside

the measured minutes cannot be excluded by design. And the estimates come from six names, one year, and a single perpetual venue in one volatility regime, so they need not carry to other venues or market states. Two candidates remain consistent with the contrast, execution urgency and the visibility of the forced fills themselves, and our design does not separate them.

Data availability

The stock-market data are licensed from a commercial vendor (Databento) and cannot be redistributed; the vendor's product identifiers, schemas, and sample windows are listed in Appendix B of the Supplementary Material so that the extracts can be reconstructed by any licensee. The blockchain records underlying the creation, redemption, liquidation, and payment-trace analyses are public on the respective chains: token issuance and redemption events are on-chain transactions on Solana, Ethereum, BNB Chain, and Arbitrum, and perpetual liquidation fills are publicly recorded on the venue's chain and preserved in a third-party archive; the acquired archive begins at the venue's listing on October 13, 2025 and extends beyond the sample; all analyses truncate it at the sample end of June 30, 2026. The extracts used here were assembled through paid access services, a Dune Analytics subscription and the requester-pays Reservoir archive, which any researcher can license; the underlying records themselves carry no redistribution restriction. The remaining public inputs include Bybit market data used for token-volume sample construction and crypto-market controls, Hyperliquid application-programming-interface data used for funding and archive cross-checks, the Solana RPC used for the payment trace, the published CBOE volatility-index close, and daily factor returns from Kenneth R. French's data library. Code that constructs the analysis panels and reproduces the tables and figures is available from the author on request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Anthropic Claude and OpenAI ChatGPT in order to assist with writing and debugging the data-processing and analysis code and with editing and proofreading the manuscript. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Table 1

Event samples.

Event sample	N	Median	P90
Large share-collateralized creations (Backed)	112	\$2,360,000	—
Liquidation fills, six names	53,962	\$360	\$5,000
Liquidation bursts (5-min $\geq$ \$250,000, merged)	39	\$430,000	\$949,000

This table reports the event samples. Large share-collateralized creations are on-chain creations of Backed tokens in the six names; the census starts at product launch in June 2025, and twelve of the 112 core-name creations precede the July 2025 start of the stock-outcome panel and enter no footprint estimate. Liquidation fills are forced-liquidation executions on the perpetual venue. Liquidation bursts aggregate fills into five-minute windows between 10:00 and 15:30 ET with total notional of at least \$250,000, merging windows within thirty minutes. Median and P90 are event sizes in dollars; the creation median is rounded to three significant digits. Stock-outcome observations run through June 2026.

Table 2

Creation events and the underlying stock.

Panel A: Localized creation test	N	Event window	Minus background	95% upper
Abnormal volume z	47	0.21* (0.10)	-0.00 (0.08)	0.39
Buy-aligned signed-flow z	47	-0.05 (0.06)	-0.06 (0.06)	0.06
Panel B: Off-hours venue test	N	Pre-30	± 5 min	Post-30
Off-hours creations: after-hours	5	0.07 (0.04)	0.08 (0.04)	0.21 (0.11)
Off-hours creations: overnight ATS	11	-0.32 (0.43)	-0.32 (0.42)	-0.44 (0.44)
Off-hours creations: premarket	30	0.28*** (0.07)	0.33*** (0.07)	0.35** (0.14)

This table reports the stock-market response to large share-collateralized tracker-token creations. Panel A is the main localized test. Each creation minute is compared with the same stock and clock minute on up to twenty nearby days, excluding other creation days. The first estimate is the mean abnormal z over the eleven one-minute observations from five minutes before the mint to five minutes after it; the second subtracts the event's own mean over minutes -30 to -6 . Standard errors clustered on the event date (a cluster-robust variance estimator with a finite-cluster correction) are in parentheses, and *, **, and *** denote significance at the 10, 5, and 1 percent levels from the clustered t -distribution with event-date degrees of freedom. The final column gives a one-sided 95% upper bound for the event-window level using t_{19} . Regular-hours volume and signed flow use XNAS.BASIC, which combines Nasdaq-family exchanges and two Nasdaq trade-reporting facilities but is not the full SIP; its median share of Nasdaq historical daily volume is 79% on these creation event days. Signed flow is classified against the contemporaneous BBO with a within-day tick-test fallback. Panel B applies the same event-time comparison to the stock venue open at each off-hours timestamp. Premarket windows are elevated before, during, and after creation, so their positive levels do not identify an event-time increase. The median creation is approximately 3%–4% of contemporaneous consolidated five-minute volume, and the USDC trace finds no matching on-chain payment for 95 covered creations. Two same-day creations in the same stock-minute define one window, leaving 47 complete windows, stamped between 10:00 and 15:30 ET, on 20 event days. Smaller or dispersed responses remain possible.

Table 3

Liquidation bursts and trading in the underlying stock.

	Volume	Aligned Flow
Panel A: Descriptive footprint		
Threshold window, abnormal z	1.68*** (0.19)	1.09*** (0.27)
Pooled difference, liquidation – creation	1.47*** (0.21)	1.14*** (0.27)
Panel B: Within-event pre-onset adjustment		
Threshold window less onset background	1.06*** (0.12)	0.93*** (0.25)
Panel C: Matched non-liquidation windows		
Pre-onset return match + background adjustment	1.08*** (0.12)	1.05*** (0.26)
Full pre-onset match, level	1.60*** (0.18)	0.97*** (0.27)
Full pre-onset match + background adjustment	1.10*** (0.12)	0.92*** (0.27)
excluding scheduled-earnings dates	1.13*** (0.15)	0.96*** (0.33)
Sign-flip p for full match + background	0.0001	0.0002

This table reports the stock-market response to forced-liquidation bursts on the perpetual venue. Events are five-minute bins between 10:00 and 15:30 ET with at least \$250,000 of forced liquidations, merging qualifying bins within thirty minutes; the rule yields 39 events on 34 dates across six stocks. Panel A reports same-stock, same-clock abnormal z -scores over the eleven one-minute observations centered on the threshold bin, the same window used for creations in Table 2. The pooled row regresses the same abnormal z on a liquidation indicator across both event classes (86 events on 53 dates) with date-clustered standard errors; it contrasts two measured responses and does not estimate an effect of liquidation. Panel B subtracts the twenty-five minutes ending five minutes before the cascade onset, defined as the first observed fill after at least thirty minutes without liquidations. Panel C compares each event with twenty clean windows in the same stock and within two hours of the same clock time. The minimal design matches the thirty-minute stock return ending before onset. The stronger design also matches pre-onset five-minute returns, realized volatility, dollar volume, absolute order imbalance, the cumulative stock return over the five trading days ending at the prior close, the tick-level stock return over the sixty seconds ending one second before onset, perpetual return and turnover, the equity-index futures return, and funding; every variable ends before the actual onset, and each control is assigned the treated event's onset-to-threshold lag. All 39 events are matched and the largest absolute paired standardized difference is 0.29. The final row excludes eight events falling on a scheduled earnings date or the preceding date. Unscheduled news remains uncontrolled, so every estimate is an association. Outcomes use XNAS.BASIC; signed flow is classified against the contemporaneous BBO. Standard errors clustered on the event date (a cluster-robust variance estimator with a finite-cluster correction) are in parentheses, and *, **, and *** denote significance at the 10, 5, and 1 percent levels from the clustered t -distribution with event-date degrees of freedom. Sign-flip tests randomize the 34 event-day means with 9,999 draws. Event-window, bin-origin, cutoff, access, and leave-one-out diagnostics appear in the Supplementary Material.

Table 3 (continued)

Liquidation bursts and trading in the underlying stock.

	Volume	Aligned Flow
Panel D: Economic scale of the full matched increment		
Mean excess dollars per event (\$ millions)	171.1*** (50.1)	31.5** (13.2)
Aggregate share of normal same-clock volume	89.2%	16.4%
Panel E: Sequencing at the quiet-period onset (35 events; 32 days)		
	Pre	Post
Aligned mid return, 2 min (bp)	46.3*** (6.4)	9.5 (6.8)
Aligned flow tilt	0.10*** (0.03)	0.09*** (0.03)
Fill-rate ratio, post / pre		1.20×** (0.07)

This table reports the economic scale of the matched increment and the sequencing of stock trading at the cascade onset. Panel D converts the full pre-onset match plus background adjustment into dollars. The percentage row divides the sum of event-level excess dollars by the sum of normal same-clock dollar volume; it is not a ratio to perpetual liquidation notional. Dollar estimates remain positive when any one stock is omitted, although the aligned-flow t -statistic falls to 2.01 in the least favorable omission. Panel E aligns stock ticks to the cascade onset, not to the later threshold-crossing bin. The onset is the first observed forced fill after at least thirty minutes without liquidations and precedes the threshold crossing in 28 of 39 events. Complete two-sided tick windows cover 35 events on 32 dates. These estimates bound the interpretation: price movement is concentrated before onset, and the post-onset increase in trading arrives without a detectable change in directional tilt. Standard errors clustered on the event date (a cluster-robust variance estimator with a finite-cluster correction) are in parentheses, with *, **, and *** denoting significance at the 10, 5, and 1 percent levels from the clustered t -distribution with event-date degrees of freedom; for the fill-rate ratio the parenthetical is the standard error of the log ratio.

Table 4

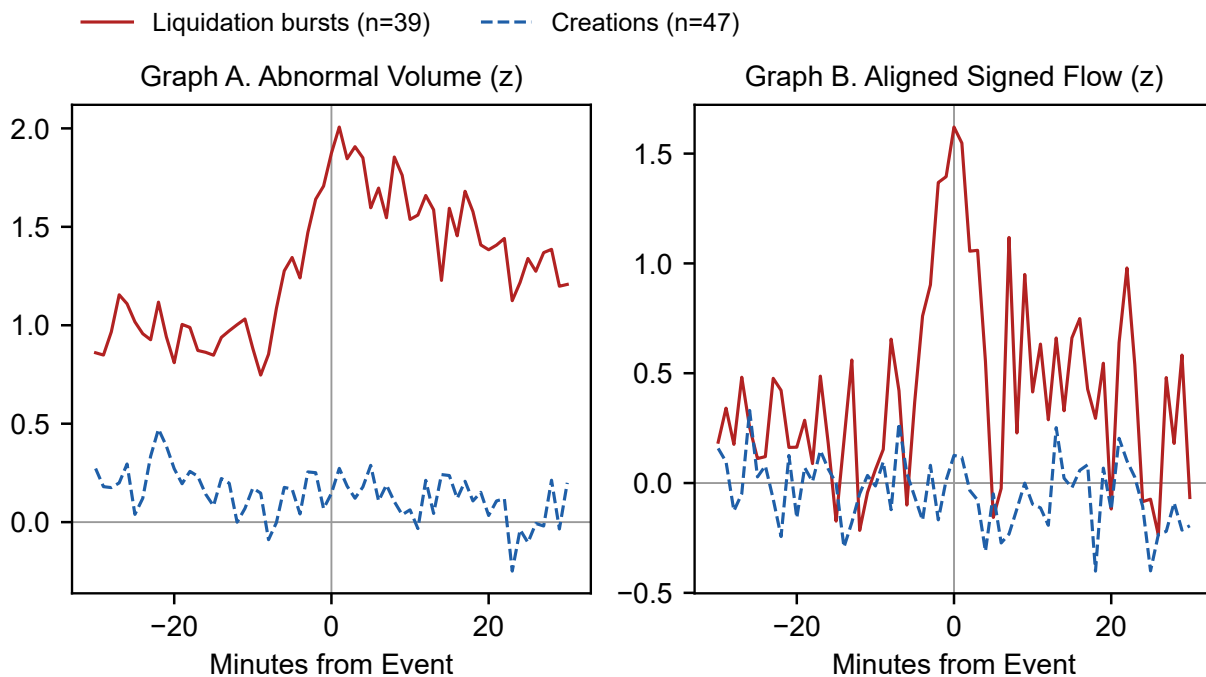
Direction and the forced–voluntary footprint contrast.

	N / days	Abnormal volume z	Aligned flow z
Panel A: Liquidation direction			
Long liquidations (falling prices)	27 / 25	1.550*** (0.211)	0.791*** (0.233)
Short liquidations (rising prices)	12 / 10	1.973*** (0.324)	1.751** (0.635)
Panel B: Forced versus voluntary pressure			
Voluntary windows	470 / 137	0.528*** (0.060)	-0.002 (0.167)
Forced – voluntary, raw	509 / 142	1.152*** (0.195)	1.088*** (0.314)
Forced – voluntary, pre-onset match	39 / 34	1.116*** (0.250)	0.775** (0.282)

This table reports the direction of the liquidation footprint and a forced–voluntary contrast. Panel A splits the 39 liquidation bursts by the direction of the forced position. Long liquidations sell into falling prices; short liquidations buy into rising prices. Stock-side outcomes use the same abnormal z constructions as Table 3, aligned with the liquidation direction, so a positive aligned flow means stock order flow tilts the same way as the forced perpetual trade. A mechanism that operates only in falling markets, such as retail margin unwinding, predicts a footprint in the long column alone. Panel B uses all-fill coverage that begins between November 12 and December 15, 2025, depending on the stock, and ends on June 30, 2026. If a stock’s first record arrives after the 10:00 ET analysis-band start, that left-censored date is removed and its sample begins on the next equity session. Voluntary windows are constructed inside actual U.S. equity sessions, accounting for holidays and early closes: the rule applies an absolute-net-taker-flow threshold of \$250,000, retains the first qualifying five-minute bin and suppresses later qualifying bins less than thirty minutes from that representative, and then excludes bins with at least \$10,000 of liquidations within thirty minutes and stock-days containing a liquidation burst. The outcome is aligned with the sign of perpetual pressure. The dollar cutoff is common, but voluntary size is absolute net taker flow whereas forced size is gross liquidation notional; among complete observations their medians are \$360,000 and \$430,000, respectively (median voluntary gross turnover is \$560,000). The rule defines 876 voluntary windows, of which 470 on 137 dates carry complete outcomes and backgrounds. The raw difference uses event-level ordinary least squares with event-date clustering. The pre-onset match pairs every forced burst with a unique voluntary window in the same stock and pressure direction and within two clock hours. Its five variables—aligned thirty-minute stock return, realized volatility, log stock dollar volume, absolute stock imbalance, and log trigger size—end before the observed forced onset or a control pseudo-onset at or before the qualifying-bin start. Outcomes do not enter the assignment. A mixed-integer assignment requires each matched-control mean to lie within 0.10 reference standard deviations of the forced mean before minimizing equal-weight standardized distance. The largest ordinary paired standardized difference is 0.097, but only 17 of 39 forced observations lie within each of the five marginal candidate ranges. Matched inference uses one pair difference and two-way clustering over 34 forced dates and 31 control dates. Date-level sign-flip p -values for the long and short direction splits in Panel A are 0.001 and 0.011 (9,999 draws). Standard errors are in parentheses; Panel A and raw Panel B rows cluster by event date, while the matched row clusters by both pair dates. *, **, and *** denote significance at the 10, 5, and 1 percent levels from the corresponding clustered t -distribution.

Fig. 1. Minute-level abnormal activity around creations and liquidation bursts

This figure shows mean abnormal activity in the underlying stock by minute relative to the event. Bursts (solid red, 39 events) are five-minute windows with forced-liquidation notional of at least \$250,000 during regular hours, overlaps merged; creations (dashed blue, 47 complete events) are large on-chain Backed creations stamped during regular hours, with same-day mints falling in the same stock-minute counted once. Both graphs use XNAS.BASIC, which consolidates Nasdaq-family exchanges and two Nasdaq trade-reporting facilities but is not the full SIP. Graph A plots abnormal log dollar volume. Graph B plots signed order flow from targeted trades with contemporaneous BBO, classified by the midpoint rule with a within-day tick-test fallback and aligned with the event direction (the liquidation's net direction for bursts, positive for creations). Abnormality is a z -score against the same stock's same minute-of-day on up to twenty control days within fifteen sessions, excluding same-type event days; both event types are benchmarked with the identical procedure within each graph. Per-minute path averages are winsorized at ± 10 standard deviations to limit the influence of occasional thin-minute z -score outliers, which affect the plotted path only and no reported statistic. Table 2 and Table 3 report the corresponding window statistics.



Supplementary Material to
“Backed or Synthetic? Execution Mechanisms and Order-Flow Footprints in the Underlying Stock Market”

This supplement collects the variable definitions, data sources, construction details, and settlement-tracing description (Appendices A–D), followed by the supplementary tables referenced in the paper (Tables IA.1–IA.9). Estimation conventions follow the main text unless stated.

Appendix A. Variable definitions

This appendix defines the variables used in the main analysis. Prices are in U.S. dollars; returns are log returns in percent unless stated otherwise. ET denotes U.S. Eastern Time.

Variable definitions

Variable	Definition
$r_{i,t}^{gap}$	Opening gap: log return from the previous official closing-auction price to day t 's official opening-auction price, from the statistics feed of the stock's primary listing exchange.
$r_{i,t}^{perp}$	Perpetual futures return over the thirty minutes ending before onset, from the trade.xyz market on Hyperliquid (ticker prefix "xyz:").
r_t^{ES}	E-mini S&P 500 futures return over the thirty minutes ending before onset, previous-tick boundary prices with a two-hour lookback.
$Mint_{i,t}$	Indicator (or log dollar size) of a primary issuance event of the share-collateralized tracker token, timestamped on chain; large events contain at least 100 shares.
$Liq_{i,t}$	Perpetual liquidation volume: forced-liquidation fills from the venue archive, aggregated to the minute or to the overnight window, signed by direction.
VIX_t	CBOE Volatility Index, daily close.
$Earn_{i,t}$	Earnings-announcement indicator from 8-K filings and the exchange calendar (Appendix B).
RV, BV, JS	Realized variance, bipower variation (Barndorff-Nielsen and Shephard, 2006), and jump share $1 - BV/RV$, from one-minute returns.

Appendix B. Data sources

This appendix lists all data sources. The stock-outcome estimation sample covers July 2025 through June 2026 for six tokenized U.S. stocks (NVDA, TSLA, CRCL, COIN, GOOGL, HOOD). The on-chain product-launch census begins in June 2025; twelve core-name June creations enter the 112-event census but precede the stock minute panel and enter no stock-footprint estimate. The perpetual contract was listed on October 13, 2025 (symbol-level start dates vary), while first all-fill records arrive between November 12 and December 15, 2025, depending on the stock, and the funding panel begins on November 12. We remove a stock’s first all-fill date when it begins after the 10:00 ET analysis-band start and begin that stock on the next equity session. The overnight trading system’s own market data is distributed from August 24, 2025, which is the vendor’s coverage start for that venue rather than a choice of ours; specifications that require these later panels begin at their stated coverage dates rather than July.

Data sources

Data	Source	Content and frequency	Used in
U.S. equities: broad-feed quotes	Databento, XNAS.BASIC	1-minute feed-level consolidated BBO (cbbo-1m), 04:00–20:00 ET; not the full SIP	Extended-session quote midpoints
U.S. equities: trades with prevailing quotes and minute volume	Databento, XNAS.BASIC publisher 93	Consolidated one-minute OHLCV and trades with the contemporaneous BBO (TCBBO), 04:00–19:59 ET; not the full SIP; daily-volume coverage medians 53%–87% across names; TCBBO extracts cover event windows and a three-minute window before each matched-control candidate (10,135 merged intervals, November 2025–June 2026)	Daily-panel outcomes, event-window volume and signed order flow, matched windows, size scaling, and second-level sequencing
U.S. equities: overnight ATS	Databento, OCEA.MEMOIR (Blue Ocean ATS)	Trade prints, 1-minute BBO and OHLCV for the 20:00–04:00 ET overnight session, the only U.S. venue trading these names in that window during the sample	Outcomes for events stamped in the 20:00–04:00 ET window
U.S. equities: opening/closing auctions	Databento, Nasdaq and NYSE auction imbalance feeds	Auction reference price, paired and imbalance quantities, 1.18M messages	Opening-auction outcomes and imbalance measures
Index futures	Databento, CME Globex (GLBX.MDP3), continuous front contracts (ES.c.0, NQ.c.0)	ES and NQ, 1-minute bars; boundary prices per Appendix C	Index-futures return in the pre-onset match

Data	Source	Content and frequency	Used in
Token trades: on-chain	Dune Analytics, <code>dex_solana.trades</code>	Solana DEX pool-level five-minute trades and volume for xStocks tokens (e.g., NVDAx), full listing history	Sample selection and token dollar volume
Token trades: exchange	Bybit public market-data API	xStocks spot pairs, five-minute klines, 9 symbols, 945k rows	Sample selection and token dollar volume
Token issuance and redemption	Dune Analytics (Solana token program; EVM ERC-20 transfers with issuer-anchor whitelists)	143,507 Solana mint/burn events (June 2025–July 2026); Ethereum 493 and BNB/Arbitrum 791 events; Ondo Ethereum 180,000 events	Creation event set
On-chain settlement tracing	Solana public RPC	Issuer operating-wallet USDC account signatures (2.73M) and transaction details around large mints	Appendix D
Perpetual futures: prices and volume	Reservoir archive of Hyperliquid (Hydromancer S3, requester-pays)	1-second bars aggregated to 5-minute and 1-hour panels; 84 markets, 1.93M 5-minute rows; cross-checked against the venue API (correlation 1.0 on overlap)	Perpetual return and turnover in the pre-onset match

Data	Source	Content and frequency	Used in
Perpetual futures: liquidations	Reservoir archive	668,943 forced-liquidation fills with price, size, direction, and account	Liquidation event set
Perpetual futures: all fills (taker-signed)	Reservoir archive	Per-fill records with the aggressing side flagged (crossed); first records range from 2025-11-12 to 2025-12-15 by stock, left-censored first dates are removed, and the analysis ends on 2026-06-30 even though the archive extends later; used to construct taker-signed net flow for the forced-versus-voluntary comparison (IA.9), excluding liquidation fills	Voluntary-burst definition and aligned-flow outcomes in Section 4.3
Funding rates	Hyperliquid API	Funding history, 67.5k rows, from November 12, 2025	Funding in the pre-onset match
Crypto market controls	Bybit public market-data API	BTC and ETH 5-minute klines (BTCUSDT, ETHUSDT), 105k rows each	Crypto-market controls
VIX	CBOE (daily close)	Volatility index level	Controls

Data	Source	Content and frequency	Used in
Earnings dates	SEC EDGAR 8-K filings; Nasdaq calendar API	Announcement dates and timing (before/after market)	Event exclusions
Dividends and calendars	Nasdaq API; exchange holiday calendar	Ex-dividend dates; trading holidays and early closes	Window construction
Off-exchange share	FINRA OTC Transparency (weekly)	Off-exchange share of consolidated volume by symbol (42–57% in our names)	Creation-settlement interpretation
Venue coverage snapshot	CoinGecko (24-hour snapshot, July 3, 2026)	Reported token volume by trading venue	Appendix C venue selection
Product legal form	Backed xStocks Product Legal Overview (docs.xstocks.fi, accessed August 4, 2026); Ondo legal disclaimer (docs.ondo.finance, accessed August 4, 2026); Ondo base prospectus approved by the Liechtenstein FMA (November 11, 2025); SEC staff Statement on Tokenized Securities (January 28, 2026)	xStocks are bearer debt instruments classified as tracker certificates; Ondo products are blockchain-based tracker certificates issued as debt instruments. The instruments provide economic exposure but are not the referenced shares or fund interests. The SEC staff taxonomy separately distinguishes security entitlements, linked securities, and security-based swaps; our share-collateralized and cash-settled labels describe execution structures rather than legal categories	Section 2

Data	Source	Content and frequency	Used in
Institutional documentation	trade[XYZ] perpetual documentation (docs.trade.xyz, accessed August 2026); issuer documentation (Backed xStocks, docs.xstocks.fi, accessed August 2026); Bybit xStocks FAQ and terms (bybit.com, accessed August 5, 2026)	External stock-derived oracle coverage and the internal fallback when external inputs are unavailable; mark-price inputs and use in margining and liquidation; issuance mechanics (market flow, in which the subscriber sends stablecoins, the broker buys the equity, and the token is delivered, with stated processing of 30 seconds up to \$200k and 2 minutes above; and in-kind delivery, in which shares are converted directly with no purchase); executing-broker order-routing practices	Sections 2 and 3

Appendix C. Price construction, windows, and filters

C.1. Token venue unit. The same Backed xStock can be deposited to and withdrawn from Bybit over supported blockchains, so the Solana DEX and Bybit observations represent one transferable issuer claim at two venues. We aggregate their recorded dollar volume but never pool different issuers' stock-linked tokens or different perpetual contracts. We exclude unverified tier-3 exchanges whose reported volume is characteristic of wash trading (Cong et al., 2023).

C.2. Boundary prices. Where a window boundary price is required, we use the previous tick: the last observed price at or before the boundary, with a maximum lookback of two hours for index futures.

C.3. Quote filters. Where quote midpoints serve as prices, we discard quotes with relative spread above 5% before computing midpoints. The 5% spread cap is stricter in our sample than the adaptive rule of Barndorff-Nielsen et al. (2009) — spreads above 50 times the same-day median — under which off-hours thresholds would be 6.9–37.3%. Trade-price series require no such filter: tightening the deviation band leaves those estimates unchanged to the reported precision.

C.4. Overnight trading in the underlying. During the sample, no exchange is open and no consolidated quotation exists between 20:00 and 04:00 ET. Residual trading in U.S. stocks over this window occurs on a single alternative trading system whose prints are reported to the tape the following morning. FINRA ATS transparency data for that system show that its weekly volume in our six names averages 0.28% to 0.78% of consolidated volume (maximum weekly share 1.8%); we therefore describe the strict window as one without a consolidated stock price rather than without any stock trading.

C.5. Official opening and closing prices. The opening gap uses each stock's official auction prices from its primary listing, not the consolidated first or last trade. We take the official opening and closing prices from the Databento statistics feed for the primary-listing exchange — Nasdaq for the five Nasdaq-listed names and NYSE for Circle — which publishes the auction-clearing price once the opening or closing cross completes. These are distinct from the pre-cross reference and imbalance prices carried on the auction-imbalance feed, which are indicative rather than executed; substituting the pre-cross reference price for the official open is not a valid measure of the opening price and changes the measured gap materially. The opening gap is the official open divided by the prior trading day's official close, minus one.

Appendix D. The USDC payment trace for primary-market settlement

For each of the 95 large mint events in the core symbols, we trace the issuer’s operating wallets on Solana, identified from a discovery query over inflow counterparties, and test whether a USDC transfer of matching size ($\pm 40\%$) precedes the mint within a $[-30, +5]$ minute window. No event shows a matching USDC leg (0 of 95). Most events have no window inflow at all (66 of 95), and the individual inflows that do occur are de minimis, with a median of \$170 against a median mint of \$2.36 million. We therefore find no evidence that these large issuances settle through an on-chain USDC payment. Issuer documentation permits in-kind creation, but in-kind delivery and off-chain cash settlement remain observationally indistinguishable; either is consistent with the absence of a public-market footprint documented in Section 4. The trace covers USDC only. The census of large Backed mints contains 138 events across eight underlyings, of which 112 are in the six core names; the trace covers the 95 of those 112 whose issuing wallets the discovery query identifies, so 17 core-name creations lie outside wallet coverage.

Supplementary material

The Supplementary Material contains: (IA.1) liquidation-burst outcome-horizon and event-definition sensitivity; (IA.2) creation-footprint robustness and family-wise inference; (IA.3) liquidation-burst timing, measurement, and market access; (IA.4) creation footprints over longer horizons; (IA.5) creations at second resolution; (IA.6) pre-onset matching balance, support, and economic magnitudes; (IA.7) opening-hour outcomes around creations and the associated power calculation; (IA.8) the trading day before a creation at thirty-minute resolution; and (IA.9) the forced-versus-voluntary comparison of perpetual pressure above a common dollar threshold.

Table IA.1

Liquidation bursts: outcome-horizon and event-definition sensitivity.

Panel A: Outcome-horizon sensitivity

Horizon (min)	Outcome	Statistic	β	t	p	Events	Dates
2	Aligned flow	Level	1.542	4.492	< 0.001	39	34
2	Aligned flow	Onset increment	1.381	4.235	< 0.001	39	34
5	Aligned flow	Level	1.086	4.073	< 0.001	39	34
5	Aligned flow	Onset increment	0.926	3.67	< 0.001	39	34
10	Aligned flow	Level	0.758	3.609	0.001	39	34
10	Aligned flow	Onset increment	0.598	2.9	0.007	39	34
15	Aligned flow	Level	0.639	3.799	< 0.001	39	34
15	Aligned flow	Onset increment	0.478	2.838	0.008	39	34
2	Volume	Level	1.815	8.446	< 0.001	39	34
2	Volume	Onset increment	1.199	7.283	< 0.001	39	34
5	Volume	Level	1.681	8.919	< 0.001	39	34
5	Volume	Onset increment	1.064	8.89	< 0.001	39	34
10	Volume	Level	1.511	8.273	< 0.001	39	34
10	Volume	Onset increment	0.894	8.791	< 0.001	39	34
15	Volume	Level	1.424	8.059	< 0.001	39	34
15	Volume	Onset increment	0.808	8.61	< 0.001	39	34

The main event definition uses a fixed \$250,000 cutoff, a calendar-aligned five-minute bin, and a thirty-minute merge gap. The first block changes only the outcome horizon while holding the 39-event sample fixed. The second shows how many observations remain under alternative cutoffs. The fixed \$250,000 rule retains all six names (39 events on 34 dates); \$100,000 retains all six with 94 events on 67 dates, while \$500,000 and \$1 million leave only 14 and 4 events. Relative cutoffs use median stock depth measured over the eligible regular-hours window, 10:00 to 15:30. They leave six CRCL events at ten percent and one event at twenty-five percent, so we report only the remaining sample counts and draw no inference from those specifications. The final block summarizes shifts in the five-minute bin origin and the merge gap. All inferential rows use date-clustered statistics and describe associations.

Table IA.1 (continued)**Panel B: Event-cutoff sensitivity**

Event rule	Events	Dates	Names	Largest-name share (%)	Inferential sample
Fixed USD 100,000	94	67	6	29.787	Yes
Fixed USD 250,000 (main)	39	34	6	38.462	Yes
Fixed USD 500,000	14	13	4	57.143	No
Fixed USD 1 million	4	4	2	75	No
10 percent of RTH depth	6	5	1	100	No
25 percent of RTH depth	1	1	1	100	No

Panel C: Bin-origin and merge-gap sensitivity

Definition change	Outcome	Statistic	β min	β max	Min $ t $	Events min	Events max
Bin origin shifts 1–4 min	Aligned flow	Level	0.943	1.189	3.942	40	42
Bin origin shifts 1–4 min	Aligned flow	Onset increment	0.768	0.983	3.29	40	42
Bin origin shifts 1–4 min	Volume	Level	1.654	1.699	8.783	40	42
Bin origin shifts 1–4 min	Volume	Onset increment	1.031	1.061	8.411	40	42
Merge gaps 15/30/60 min	Aligned flow	Level	1.045	1.086	4.073	39	41
Merge gaps 15/30/60 min	Aligned flow	Onset increment	0.886	0.926	3.67	39	41
Merge gaps 15/30/60 min	Volume	Level	1.681	1.681	8.919	39	41
Merge gaps 15/30/60 min	Volume	Onset increment	1.064	1.064	8.89	39	41
Cutoff USD 100k (94 events, 67 dates)	Aligned flow	Level	1.123	1.123	8.135	94	94
Cutoff USD 100k (94 events, 67 dates)	Aligned flow	Onset increment	0.912	0.912	6.922	94	94
Cutoff USD 100k (94 events, 67 dates)	Volume	Level	1.571	1.571	13.783	94	94
Cutoff USD 100k (94 events, 67 dates)	Volume	Onset increment	1.16	1.16	14.359	94	94

Table IA.2

Creation-footprint robustness and family-wise inference.

Panel A: Off-hours creation footprints

Session	Events / dates	Pre-30 (SE)	Event window (SE)	Post-30 (SE)
After-hours	5 / 3	0.075 (0.037)	0.078 (0.039)	0.211 (0.105)
Overnight	11 / 8	-0.320 (0.427)	-0.322 (0.424)	-0.444 (0.440)
Premarket	30 / 11	0.282 (0.073)	0.333 (0.075)	0.349 (0.139)

Panel B: Family-wise inference

Outcome	t	Raw permutation p	Max- t adjusted p
log_vol	-0.32	0.758	1
log_rv	-0.39	0.695	0.999
log_bv	-0.97	0.303	0.949
jump_sh	2.14	0.043	0.232
espread_bp	-0.96	0.447	0.951
abs_auc_imb	-0.52	0.594	0.998
abs_gap_bp	0.34	0.751	1
sflow_norm	-1.27	0.223	0.799

The first block reports minute-level abnormal volume around creation timestamps outside regular hours, measured on the venue open at the time (extended-session volume after hours and in the premarket; overnight-system volume at night). The four-hour TWAP test shows no accumulation of buying around creations. The pre-event four-hour window has a small sell tilt (-0.11 , $t = -3.3$), a pattern that does not resemble concentrated issuer buying. The second block reports the family-wise max- t adjustment across the eight opening-hour outcomes in Table IA.7.

Table IA.3

Liquidation-burst timing, measurement, and market access.

Panel A: Cascade-onset sequencing

Statistic	Mean	t
Aligned return, 2 min before onset (bp)	46.301	7.18
Aligned return, 2 min after onset (bp)	9.542	1.4
Aligned return, 5 min before onset (bp)	76.482	6.43
Aligned return, 5 min after onset (bp)	21.717	1.88
flow tilt, 2 min before onset	0.098	3.22
flow tilt, 2 min after onset	0.085	2.96
tilt step at onset	-0.013	-0.45
fill-rate ratio, after / before onset (geo. mean)	1.204	2.73

Panel B: Background-clock sensitivity

Outcome	Statistic	β	t	p	N / days	LOO min $ t $
Abnormal volume z	Less pre-onset background	1.064	8.89	< 0.001	39 / 34	
Abnormal volume z	Less threshold-bin background	0.717	6.02	< 0.001	39 / 34	
Aligned flow z	Less pre-onset background	0.926	3.67	< 0.001	39 / 34	3.44
Aligned flow z	Less threshold-bin background	0.867	3.78	< 0.001	39 / 34	3.52

Volume comes from XNAS.BASIC minute bars. Signed order flow uses trades classified against the contemporaneous BBO with a within-day tick-test fallback. The onset-background rows correspond to the main estimates; the threshold-background rows show how using the later event-definition clock changes the comparison.

The first block re-times each burst at the operational cascade onset, the first observed fill after at least thirty minutes without liquidations, and reports aligned returns, flow tilts, and the ratio of fill arrival rates over the two minutes on each side. The fill-rate ratio is summarized as a geometric mean and its t is computed on day-level logs. The second block isolates the only background-clock sensitivity relevant to the main estimates: a background ending before the cascade onset versus one ending before the later threshold bin. The latter subtracts early forced fills in 28 of 39 events and is therefore a conservative diagnostic. The leave-one-out column is available for signed flow; IA.6 reports the corresponding dollar estimates when each stock is omitted in turn. The remaining blocks audit trade classification and report stock-side venue footprints by session: extended-session volume for after-hours and premarket bursts and overnight-system volume for night bursts. Weekend bursts have no open stock venue and therefore no contemporaneous stock outcome. Main-table estimates and creation comparisons are not repeated.

Table IA.3 (continued)**Panel C: Trade-classification data quality**

Quality check	Value
Raw trade-and-quote records	177,421,705
Midpoint / tick-test classified	86.6663% / 13.3334%
Unresolved trades	455 (0.000256%)
Trade-file / minute-bar mismatched minutes	0

Panel D: Stock-market access by session

Session	Events / days	Level z (SE)	Less threshold background (SE)
After-hours	28 / 20	1.515 (0.174)	0.563 (0.147)
Premarket	20 / 16	1.255 (0.178)	0.481 (0.147)
Overnight	18 / 16	1.194 (0.152)	0.508 (0.092)

At the same \$100,000 event cutoff, a separate access-design construction, which uses the full 9:30 to 16:00 session and merges qualifying bins across session boundaries, counts 159 regular-hours events and 5 weekend events; the main pipeline, which restricts to 10:00 to 15:30 ET before merging, yields the 94 events reported in Table IA.1. Weekend events have no contemporaneous stock outcome because no stock venue is open.

Table IA.4

Creation footprints over longer horizons.

Benchmark	CAR[1,5]	CAR[1,10]	CAR[1,20]
Market model, Nasdaq-100 (QQQ)	1.30 (0.53)	1.92 (0.80)	2.44 (1.40)
Two-factor, QQQ + bitcoin	1.35 (0.56)	1.95 (0.84)	2.39 (1.49)
Market model, S&P 500 (SPY)	1.08 (0.54)	1.50 (0.95)	2.00 (1.61)
Fama–French five-factor	0.87 (0.59)	1.22 (0.90)	0.63 (1.54)
Market-adjusted (beta = 1, QQQ)	0.12 (0.67)	-0.49 (0.91)	-3.08 (1.71)
Cross-sectional mean (six names)	-0.20 (0.33)	-0.65 (0.56)	-0.80 (0.95)

Cumulative abnormal returns to large share-collateralized tracker-token creations over one, two, and four trading weeks, aligned to the first regular-session open after each on-chain creation (96 creation stock-days on 37 event dates). Each row uses a different benchmark for the abnormal return: two market models (Nasdaq-100 and S&P 500 exchange-traded funds), a two-factor model that adds the bitcoin return, the Fama–French five-factor model, a beta-of-one market adjustment, and the cross-sectional mean of the six sample stocks. Loadings are estimated over the full sample; each cell is the mean event-level cumulative abnormal return in percent with a date-clustered standard error in parentheses. Abnormal trading volume over the same windows, measured against each stock’s own trailing baseline and therefore common to all benchmarks, is -6.6 percent ($t = -0.80$) over the first week and -5.5 percent ($t = -1.06$) over four weeks. The volume estimate is indistinguishable from zero under every benchmark. Return estimates range from -3.08 to $+2.44$ percent and are significant only under the Nasdaq-100 models, where the estimate is concentrated in the three crypto-linked names with the lowest market-model R^2 (0.16 to 0.32). Because the broad benchmarks may not fully absorb the crypto and growth factors with which creations cluster, we treat the mid-horizon return estimates as benchmark-dependent rather than as creation-specific and do not infer a directional longer-horizon price footprint.

Table IA.5

Creations at second resolution.

Statistic	Mean	t	One-sided 95	
Buy-aligned flow tilt, 2 min before	0.01	0.45	0.05	20
Buy-aligned flow tilt, 2 min after	-0.03	-1.09	0.018	20
Tilt step at the timestamp	-0.041	-1.32	0.013	20
Fill-rate ratio, after / before (geo. mean)	0.989	-0.23	1.079	20
Aligned mid return, 2 min before (bp)	2.556	1.08	6.655	20
Aligned mid return, 2 min after (bp)	5.092	2.45	8.686	20
Aligned mid return, 5 min before (bp)	0.71	0.23	6.155	20
Aligned mid return, 5 min after (bp)	3.898	0.96	10.898	20

The second-level design of Section 5.3 applied at the creation timestamp, so that the two event classes are compared on the same clock. Events are the 45 regular-hours creation windows with a complete tick sample; the alignment direction is a purchase, since the hypothesized footprint is the issuer's market buy, so a positive tilt would support the channel. Statistics are the flow tilt and the fill arrival rate over the two minutes on each side of the timestamp, with the aligned mid return shown at two- and five-minute windows; trades are signed against the contemporaneous best quote, as in the liquidation exercise. Inference clusters by event date; one-sided ninety-five percent bounds use a t critical value at the number of event dates less one, and the fill-rate ratio is summarized as a geometric mean with its bound on the day-level logs. The flow and fill-rate estimates are indistinguishable from zero. The aligned mid return over the two minutes after the timestamp is positive (5.1 basis points, $t = 2.45$); it is one of eight statistics reported here and is not accompanied by any change in directional flow or fill arrival. The bound on the post-event tilt, 0.018, and on the fill-rate ratio, 1.08, both lie below the corresponding liquidation values of 0.10 and 1.20 reported in Table 3, so the creation null is not an artifact of measuring at the minute.

Table IA.6

Pre-onset matching: balance, support, and economic magnitudes.

Panel A: Pre-onset covariate balance

Design	Pre-onset covariate	Treated mean	Control mean	Paired SMD	Events
minimal_pre_return	ret30_bp	-29.841	-29.338	-0.005	39
strong_pre_onset	abs_flow25	0.108	0.104	0.036	39
strong_pre_onset	es_ret30_bp	-1.566	-3.661	0.093	39
strong_pre_onset	fund_bph	0.158	0.149	0.044	39
strong_pre_onset	log_perp_turn30	14.017	13.716	0.291	39
strong_pre_onset	logvol25	15.568	15.398	0.108	39
strong_pre_onset	perp_ret30_bp	-46.638	-34.366	-0.083	39
strong_pre_onset	r5_bp	-25.38	2.802	-0.021	39
strong_pre_onset	ret30_bp	-29.841	-29.13	-0.007	39
strong_pre_onset	ret5_bp	2.875	-1.289	0.056	39
strong_pre_onset	ret60s_bp	4.124	1.046	0.107	39
strong_pre_onset	rv30_bp	119.106	102.762	0.222	39

This section reports diagnostics that are not repeated in Table 3. Candidate controls are clean windows in the same stock, within two hours of the same clock time, with less than \$10,000 of liquidations in the surrounding thirty minutes. We assign each control the treated event's onset-to-threshold lag. Matching variables are measured before the resulting pseudo-onset, and outcomes are measured around the corresponding pseudo-threshold. The minimal match uses only the sign and nearest magnitude of the stock's thirty-minute pre-onset return. The stronger match adds the five-minute stock return, realized volatility, dollar volume, absolute order imbalance, the cumulative stock return over the five trading days ending at the prior close, the tick-level stock return over the sixty seconds ending one second before onset, perpetual return and turnover, equity-index-futures return, and funding. All 39 bursts obtain twenty controls. We do not measure matching variables in the threshold bin because onset precedes that bin in 28 events and the bin may therefore contain post-onset information. The first table reports paired standardized differences for all pre-onset covariates. The second converts the fully matched and background-adjusted differences into dollars and reports date-clustered and sign-flip inference. The third reports ranges obtained by omitting one stock at a time. Table 3 excludes scheduled earnings dates and the preceding dates; unscheduled news remains uncontrolled. These amounts describe conditional associations, not dollars caused per dollar of liquidation.

Table IA.6 (continued)**Panel B: Economic magnitudes and inference**

Outcome	Mean excess (USD mn)	t	p	Sign-flip p	Events / days
Abnormal dollar volume	171.123	3.417	0.002	<0.001	39 / 34
Liquidation-aligned abnormal signed dollar flow	31.486	2.388	0.023	0.001	39 / 34

Panel C: Relative scale, leave-one-stock-out ranges, and match support

Outcome	Aggregate % of normal volume	Leave-one-stock-out range (USD mn)	Leave-one-stock-out min $ t $
Abnormal dollar volume	89.212	55.9–256.0	3.167
Liquidation-aligned abnormal signed dollar flow	16.415	17.6–46.8	2.01

All 39 bursts are matched; the smallest clean candidate pool contains 1,337 windows. Duplicate stock–threshold bins, stock minutes, and perpetual bins are all zero.

Table IA.7

Opening-hour outcomes around creations, and why they carry no power.

Outcome	Backed	t	Wild p	Ondo	t
log_vol	-0.019	-0.321	0.762	0.015	1.125
log_rv	-0.027	-0.387	0.703	0.024	1.523
log_bv	-0.069	-0.967	0.352	0.02	1.187
jump_sh	0.026	2.14	0.043	0.003	1.102
espread_bp	-0.534	-0.962	0.405	0.013	0.247
abs_auc_imb	-0.012	-0.522	0.624	-0	-0.025
abs_gap_bp	8.102	0.344	0.806	5.334	0.946
sflow_norm	-0.017	-1.27	0.222	0.001	0.644

This table reports the session-level tests that Section 4 mentions but does not tabulate. Opening-hour outcomes are regressed on an indicator for a creation aligned to the first open after its timestamp and on the second issuer's standardized daily primary flow, with lagged stock variance, VIX, and bitcoin variance controls, stock and month fixed effects, and date-clustered standard errors; the wild-bootstrap p uses 999 draws. The eight outcomes were chosen before estimation to span the forms a share purchase could take: dollar volume for quantity, realized variance for total price movement, bipower variation for the continuous part that a worked order would produce, the jump share for the discontinuous part that a block would produce, effective spreads for the cost of consuming liquidity, auction imbalance for execution in the opening cross, the absolute gap for a move in the price level, and signed order flow for direction. Dollar volume and signed order flow are the primary outcomes, since the hypothesized channel is a flow of orders, and the remaining six are secondary checks. Nothing responds, and the one nominally significant coefficient has a max- t adjusted permutation p -value of 0.23 across the eight. A joint test that both issuers' flow variables are zero does not reject, at $p = 0.90$ for opening volume and $p = 0.20$ for realized variance, and the two are essentially orthogonal in the panel, with a correlation of -0.002 . We report these results but do not read them as evidence. The outcomes are measured over the opening hour, in which the median stock trades \$1.06 billion, while a creation of median size is \$2.09 million, or 0.20 percent of that hour's volume. The standard error on signed order flow is 1.31 percent of volume, some seven times the quantity at issue, so a purchase of that size executed entirely within the hour would move the coefficient by about a seventh of one standard error. The second issuer's creations are smaller still, at a median of about sixty shares. Power in this setting comes from measuring over minutes rather than over the opening hour, which is what Table 2 does.

Table IA.8

The trading day before a creation, at thirty-minute resolution.

Panel A: Intervals 26:00 to 20:00 hours before creation

Hours before creation	Buy-aligned flow t	Volume t	Buy-aligned flow t (day)	Volume t (day)	Buy-aligned flow z	Volume z
26:00	-0.669	0.212	-0.11	-0.639	-0.263	0.041
25:30	-0.613	0.909	-0.987	-0.002	-0.208	0.363
25:00	-0.3	0.287	1.028	-0.451	-0.083	0.084
24:30	-0.124	0.39	0.327	0.022	-0.034	0.095
24:00	-0.701	-0.509	0.136	-0.529	-0.143	-0.078
23:30	0.562	0.656	1.357	1.521	0.094	0.162
23:00	0.392	1.96	1.03	1.036	0.087	0.345
22:30	0.59	1.291	0.425	1.033	0.087	0.267
22:00	0.968	0.952	0.752	1.074	0.102	0.189
21:30	-0.328	0.469	-0.737	0.491	-0.04	0.097
21:00	-0.004	0.474	1.057	1.098	-0.001	0.097
20:30	-2.33	0.845	-2.066	1.048	-0.387	0.129
20:00	0.238	1.522	0.433	1.754	0.032	0.315

A custody-backed issuer that acquires shares before it mints would trade ahead of the on-chain timestamp, outside the window that Table 2 centres on the mint. This table walks backwards through the observable trading minutes before each regular-hours creation—from thirty minutes before the mint back through the same day’s 04:00 open, and then through the previous trading day from 19:30 to 04:00—in thirty-minute intervals; the interval index counts trading minutes, not calendar time, so a Monday mint is compared with the preceding Friday rather than with the closed weekend. Each interval is compared with the same stock’s same clock interval on up to twenty control days, excluding other creation days, exactly as in Table 2. Entries are mean abnormal z with date-clustered t -statistics on event-level observations; the day-weighted variant collapses each event date to one observation. The feed does not observe 20:00 to 04:00, so overnight purchases are outside this test, as are purchases spread thinly across the day. Buy-aligned flow shows no consistent direction: the intervals that cross $|t| = 2$ under event-level clustering are as often negative as positive, only one survives in the day-weighted variant, and the mean abnormal flow across all intervals is -0.01 . Abnormal volume is mildly elevated on average without carrying direction. Concentrated single-interval purchasing during observable hours would appear here; we cannot exclude overnight or evenly spread execution.

Table IA.8 (continued)**Panel B: Intervals 19:30 to 13:30 hours before creation**

Hours before creation	Buy-aligned flow t	Volume t	Buy-aligned flow t (day)	Volume t (day)	Buy-aligned flow z	Volume z
19:30	0.591	1.797	-1.17	1.125	0.086	0.252
19:00	0.731	0.547	0.471	0.609	0.153	0.095
18:30	-1.061	0.847	-1.231	1.646	-0.162	0.161
18:00	0.818	2.471	0.772	2.359	0.116	0.516
17:30	-1.154	1.541	-1.81	1.827	-0.163	0.265
17:00	0.094	0.892	0.337	1.684	0.012	0.142
16:30	-0.263	0.962	0.145	1.461	-0.039	0.175
16:00	-1.215	1.87	-0.726	1.676	-0.143	0.298
15:30	-1.199	1.568	0.576	0.87	-0.137	0.256
15:00	0.566	0.689	0.538	0.866	0.081	0.117
14:30	-0.576	2.645	0.416	1.853	-0.109	0.359
14:00	-0.227	2.167	0.958	1.56	-0.026	0.327
13:30	2.323	1.271	0.979	1.914	0.208	0.2

Table IA.8 (continued)**Panel C: Intervals 13:00 to 7:00 hours before creation**

Hours before creation	Buy-aligned flow t	Volume t	Buy-aligned flow t (day)	Volume t (day)	Buy-aligned flow z	Volume z
13:00	1.394	2.273	1.244	2.15	0.211	0.297
12:30	2.497	1.534	1.646	1.188	0.274	0.331
12:00	0.864	0.409	1.341	0.286	0.093	0.07
11:30	-0.364	1.291	-0.233	1.343	-0.044	0.171
11:00	-0.946	0.368	-1.232	0.219	-0.126	0.054
10:30	-1.751	0.431	-0.994	0.329	-0.26	0.089
10:00	-0.502	1.322	-0.378	1.622	-0.077	0.259
9:30	-0.746	2.26	-0.67	1.688	-0.182	0.466
9:00	-2.02	1.188	-1.225	0.699	-0.32	0.352
8:30	-0.027	1.793	-0.714	1.57	-0.004	0.46
8:00	1.843	0.508	0.214	0.378	0.228	0.152
7:30	-0.768	2.367	-0.794	2.979	-0.187	0.536
7:00	-1.254	0.98	-0.815	0.954	-0.198	0.208

Table IA.8 (continued)**Panel D: Intervals 6:30 to 0:30 hours before creation**

Hours before creation	Buy-aligned flow t	Volume t	Buy-aligned flow t (day)	Volume t (day)	Buy-aligned flow z	Volume z
6:30	0.141	2.591	-0.491	1.22	0.034	0.528
6:00	0.169	1.623	-0.786	0.65	0.029	0.311
5:30	-0.22	1.789	-0.821	0.995	-0.057	0.435
5:00	-0.163	0.329	-0.67	-0.852	-0.028	0.069
4:30	2.461	1.106	1.318	0.52	0.379	0.234
4:00	-1.357	0.479	-1.514	0.398	-0.218	0.074
3:30	0.158	1.351	0.84	1.407	0.037	0.175
3:00	0.877	0.97	0.499	0.973	0.118	0.175
2:30	2.008	1.473	1.515	1.754	0.377	0.221
2:00	1.391	1.194	0.728	0.921	0.297	0.146
1:30	-0.553	0.991	-0.745	1.82	-0.078	0.178
1:00	0.011	2	-0.061	2.719	0.001	0.328
0:30	1.349	1.966	0.767	2.221	0.196	0.226

Table IA.9

Forced and voluntary perpetual-pressure windows.

Panel A: Raw fixed-threshold and percentile comparisons

Sample / contrast	Outcome	Estimate	SE	<i>t</i>	<i>p</i>	N	Days
Voluntary mean, fixed threshold	Abnormal volume z	0.528	0.06	8.855	<0.001	470	137
Forced minus voluntary, fixed threshold	Abnormal volume z	1.152	0.195	5.917	<0.001	509	142
Voluntary mean, fixed threshold	Pressure-aligned flow z	-0.002	0.167	-0.01	0.992	470	137
Forced minus voluntary, fixed threshold	Pressure-aligned flow z	1.088	0.314	3.46	<0.001	509	142
Forced minus voluntary, top one percent	Abnormal volume z	1.15	0.193	5.97	<0.001	365	123
Forced minus voluntary, top one percent	Pressure-aligned flow z	1.01	0.282	3.58	<0.001	365	123

This section reports the forced–voluntary comparison referenced in Section 4.3. The venue’s fill records mark the aggressing side of every trade, so voluntary pressure is measured rather than inferred. First records arrive between November 12 and December 15, 2025, depending on the stock. If the first record arrives after the 10:00 ET analysis-band start, we remove that left-censored date and begin on the next equity session; every stock’s analysis ends on June 30, 2026. We first restrict bins to actual U.S. equity sessions, including holiday closures and early closes; apply an absolute net-taker-flow threshold of \$250,000; retain the first qualifying five-minute bin and suppress later qualifying bins less than thirty minutes from that representative; and then exclude windows with at least \$10,000 of liquidations within thirty minutes and stock-days containing a liquidation burst. Liquidation fills themselves are excluded from voluntary signed flow. This sequence defines 876 windows through June 30, 2026, of which 470 on 137 dates carry complete eleven-minute outcomes and twenty-five-minute backgrounds. The dollar threshold is shared, but voluntary size is absolute net taker flow whereas forced size is gross liquidation notional; they are matching dimensions, not identical economic objects. Among bins with nonzero net pressure in the full available five-minute panel, the within-stock correlation between log absolute net pressure and log gross turnover ranges from 0.817 to 0.883, so the net-pressure rule also selects active windows. Panel A reports the raw fixed-threshold comparison and a top-one-percent threshold sensitivity. Raw differences use event-level ordinary least squares with event-date clustering. Panel B reports the one-to-one pre-onset match and a sensitivity that subtracts separately measured pre-onset backgrounds. For every forced burst, the candidate set is restricted to the same stock, pressure direction, and a clock time within two hours. Forced states end before the observed onset. The treated lead is the ceiling, in minutes, of the interval from onset to the first fill in its qualifying bin; subtracting that nonnegative lead from a candidate’s bin start defines the pseudo-onset and ensures that its state ends before the qualifying bin. The five variables are aligned thirty-minute stock return, realized volatility, log stock dollar volume, absolute stock imbalance, and log trigger size. Log volume is the mean of log one plus one-minute dollar volume over thirty minutes; absolute imbalance is the absolute sum of signed stock flow divided by total stock dollar volume over those minutes. The background sensitivity subtracts the twenty-five-minute mean ending five minutes before the actual or pseudo-onset. Outcomes never enter the assignment. A mixed-integer assignment requires the matched-control mean of every variable to lie within 0.10 reference standard deviations of the forced mean and then minimizes equal-weight standardized distance, using each control once. Matched inference uses one difference per pair and two-way clustering by forced and control date. Panels C and D report balance and support. Although observed mean balance is close, candidate-range overlap is incomplete; the estimates are conditional associations and do not separate urgency, forced-fill visibility, remaining state selection, or common shocks.

Table IA.9 (continued)**Panel B: Balance-constrained one-to-one pre-onset match**

Sample / contrast	Outcome	Estimate	SE	<i>t</i>	<i>p</i>	N	Days
Forced minus matched voluntary	Abnormal volume z	1.116	0.25	4.468	<0.001	39	34
Forced minus matched voluntary	Pressure-aligned flow z	0.775	0.282	2.748	0.010	39	34
Forced minus matched voluntary	Abnormal volume z, less pre-onset background	0.825	0.178	4.638	<0.001	39	34
Forced minus matched voluntary	Aligned flow z, less pre-onset background	0.762	0.252	3.019	0.005	39	34

Panel C: Pre-onset covariate balance

Pre-onset variable	Forced mean	Control mean	Paired SMD	Reference scale	Pairs
Aligned 30-minute stock return	108.128	99.798	0.085	102.682	39
30-minute realized volatility	119.204	114.072	0.072	73.277	39
Log 30-minute stock dollar volume	15.556	15.61	-0.032	1.647	39
Absolute 30-minute stock imbalance	0.108	0.099	0.076	0.096	39
Log trigger size	13.056	13.002	0.097	0.538	39

Panel D: Candidate-pool and range-support diagnostics

Diagnostic	Value
Eligible assignment edges	893
Candidate-pool size, minimum	9
Candidate-pool size, median	21
Candidate-pool size, maximum	46
Largest absolute paired SMD	0.097
Inside candidate range: aligned return	34
Inside candidate range: realized volatility	30
Inside candidate range: log stock volume	35
Inside candidate range: absolute imbalance	35
Inside candidate range: log trigger size	34
Inside each of five marginal candidate ranges	17